CausalPose-3D: An Automated 3D Scene Reconstruction-Driven Teleoperation Assistance System for Humanoid Bimanual Manipulation

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***Abstract*—Remote human-in-the-loop teleoperation of humanoid bimanual manipulation tasks—such as grasping a prismatic beverage carton and pouring liquid into a narrow cylindrical cup—is severely bottlenecked by multi-body visual self-occlusion, cross-camera viewpoint discontinuity, and monocular perspective depth-parallax ambiguity when operators rely on conventional 2D camera streams. While automated 3D scene reconstruction offers a principled foundation for spatial teleoperation assistance, existing monocular depth foundation models, feed-forward multi-view pointmap transformers, and unconstrained generative mesh completion networks either exhibit decimeter-scale metric drift or hallucinate unverified geometry in safety-critical workspaces. In this paper, we present CausalPose-3D (*Causal Observability-Conditioned 6D Pose Tracking and 3D Surface Reconstruction*), an end-to-end teleoperation assistance framework and system driven by automated, sensor-grounded 3D scene reconstruction on a 61-link humanoid mobile manipulator (16 active torso, head, and dual-arm degrees of freedom plus parallel-jaw grippers) equipped with four heterogeneous RGB-D sensors (head, chest, left-wrist, and right-wrist cameras). We first conduct a systematic empirical benchmark of 31 candidate perception, depth-estimation, feature-matching, point-tracking, and 3D reconstruction frameworks (including Prior-DA, Depth Pro, Depth Anything V2, UniDepthV2, VGGT, SLAM3R, DUSt3R, MASt3R, CUT3R, Florence-2, Grounding DINO, SAM 2.1, YOLO11-seg, BundleTrack, BundleSDF, FoundationPose, Any6D, BIT, 6DOPE-GS, KV-Tracker, Point2Pose, Online BootsTAPIR, LoFTR, Efficient-LoFTR, XFeat, ALIKED+LightGlue, and RoMa), establishing why sensor-anchored multi-camera geometric fusion outperforms standalone neural depth or pointmap inference and formalizing why increasing pose availability or lowering internal factor-graph cost can degrade strict pose accuracy and accumulated 3D surfaces under degenerate depth support. Building on these findings, our CausalPose-3D reconstruction engine synthesizes four core technical pillars to power real-time operator assistance: (1) *MoCap-free bilateral kinematic-visual extrinsic closure* on $\mathrm{SE}(3)^3$ across $1{,}648$ training frames ($12{,}061$ observations) and 413 held-out test frames ($2{,}690$ observations) evaluated without test-split scene refitting, achieving $2.52\,\mathrm{mm}$ (left arm) and $6.56\,\mathrm{mm}$ (right arm) median 3D tabletop residuals; (2) *automated 3D functional workspace reconstruction and causal pose/surface governance* in the metric tabletop frame O_{table} ($251{,}182$ depth inliers across $0.92 \times 0.49\,\mathrm{m}$), combining watertight parametric revolution completion of the target receptacle ($\varnothing_{\mathrm{top}} = 85.4\,\mathrm{mm}$, cross-camera rim agreement $\leq 1.60\,\mathrm{mm}$) with causal multi-view surfel accumulation ($5{,}699$ surfels), deterministic CUDA RANSAC and exact $\mathrm{SE}(3)$ inverse-pose transfer (raising public Laval Clock/Jar strict successes from $135/107$ to $191/136$), guarded `NO_BA` non-keyframe retirement (saving 928.4–$1{,}994.0\,\mathrm{MiB}$ RSS), array-native `support10/coarse12mm` $1{,}500$-point budget selection (raising public C_{10} from $40.57\%/52.34\%$ to $85.10\%/85.05\%$ and task surface coverage from 71.78% to 89.97% while cutting selector latency by 33%), and initial upright bounding-prism completion ($67.8 \times 94.8 \times 199.5\,\mathrm{mm}$) across $498/498$ composite frames; (3) *projective* $\mathbb{RP}^2$ *unoriented aperture geometry* (absorbing all 11 observed 180° SVD normal flips across $1{,}992$ camera frames with $0.000\,\mathrm{mm}$ task-metric drift) paired with *role-isolated SAM 2 video segmentation memory, ViT feature caching, same-thread SAM 2 GPU-window CPU preparation overlap, and request-scoped $O(\textit{runs})$ integer RLE evidence deduplication* (reducing four-camera full-chain p_{95} latency by 21.7% from $549.15\,\mathrm{ms}$ to $429.94\,\mathrm{ms}$ and cutting warm configured pouring HTTP endpoint latency by 27.6% from $176.40\,\mathrm{ms}$ [$155.33/219.06\,\mathrm{ms}$ p_{50}/p_{95}] to $127.80/153.40\,\mathrm{ms}$, cutting exclusive outer object-world wrapper overhead by 72.9% from 23.17 to $6.28\,\mathrm{ms}$, and eliminating $140.58\,\mathrm{ms}$ mid-request Generation-2 GC stalls with 100% bit-exact parity across 499 online equivalence checks); and (4) an *orthographic spatial teleoperation assistance interface* providing instantaneous ($0\,\mathrm{ms}$) canonical top-down (XY planar bullseye) and side-elevation (XZ vertical clearance) projections alongside a real-time vector bullseye HUD. Across the $37.47\,\mathrm{s}$ (498-frame) multi-camera manipulation study, CausalPose-3D verifies $11.05\,\mathrm{mm}$ physical grasp-closure accuracy and guides continuous pouring alignment across a 124-frame ($9.5\,\mathrm{s}$) bullseye window, reaching a peak radial safety margin of $+32.8\,\mathrm{mm}$ inside the $42.7\,\mathrm{mm}$ cup rim ($9.8\,\mathrm{mm}$ radial offset at 63.2° tilt).**

***Index Terms*—CausalPose-3D, teleoperation assistance system, automated 3D scene reconstruction, humanoid bimanual manipulation, hand-eye kinematic calibration, foundation model benchmark, projective $\mathbb{RP}^2$ aperture geometry, orthographic spatial interface.**

I. Introduction

REMOTE human-in-the-loop teleoperation remains the foundational paradigm for deploying humanoid and bimanual mobile manipulators in unstructured physical environments—both for direct supervisory execution of high-consequence tasks and for collecting high-precision demonstrations to train visuomotor imitation policies [1]–[6], [13]–[15]. Despite rapid progress in robot actuation and whole-body controllers, the operational success rate of remote bimanual manipulation is fundamentally bottlenecked by the *visual perception gap* between the remote physical workspace and the human operator [1], [2], [7]–[12].

The overarching objective of this work is to build an **automated 3D scene reconstruction-driven teleoperation assistance system**: a framework that continuously and automatically reconstructs the metric 3D state of the manipulation workspace—including the physical support tabletop, stationary target containers, dynamically manipulated objects, and the

The authors are with Tiwater Research, Tiwater (https://github.com/tiwater). Project page: https://tiwater.github.io/causalpose3d/.

full 61-link robot embodiment—directly from onboard multi-camera RGB-D streams without external motion capture (MoCap) or pre-loaded object CAD models, and transforms that live 3D reconstruction into quantitative, parallax-free spatial guidance for the teleoperator (Fig. 1).

Consider a representative bimanual liquid-pouring task executed on our 61-link humanoid manipulator: the operator must guide the right arm to grasp an upright beverage carton resting on the tabletop, lift and transport it toward a cylindrical receiving cup, and progressively tilt the carton up to 80° so that the pouring spout $\mathbf{p}_{\text{spout}}(t)$ remains strictly inside the vertical cylinder above the target cup aperture centered at $\mathbf{p}_{\text{cup}}$ while maintaining positive vertical rim clearance $\Delta z(t) > 0$. When an operator relies on conventional 2D video streams from onboard RGB-D cameras, three compounding perceptual failure modes arise:

1) **Severe Multi-Body Self-Occlusion:** As the right arm grasps and tilts the beverage carton, the distal forearm links, parallel-jaw gripper fingers, and the upper body of the carton itself occlude the target cup rim in the chest and wrist camera views.
2) **Projective Depth-Parallax Ambiguity:** Under a perspective pinhole camera model $\pi_c(\mathbf{X}) = \mathbf{K}_c\mathbf{X}/X_z$, a pouring spout located at elevation Δz above the cup rim but displaced by Δx along the viewing ray projects to the exact same 2D pixel coordinates as the cup center whenever $\Delta x/\Delta z = \cot\alpha_c$, where α_c is the camera viewing elevation angle [7], [85]. Consequently, an operator viewing a 2D perspective monitor cannot reliably discern whether the spout is centered over the cup opening or 40–$60\,\text{mm}$ behind the rear rim.
3) **Fragmented Single-View Depth Shells vs. Generative Hallucination:** Rendering raw single-frame RGB-D point clouds or foreground masks in a 3D viewer produces an incomplete, paper-thin front surface shell of the held object that lacks back walls, side depth, and tabletop support grounding [11]. Conversely, applying unconstrained monocular depth networks [16]–[23], feed-forward pointmap transformers [24]–[30], or generative shape completion [77]–[79] risks inventing plausible-looking but metrically false geometry, violating the core safety requirement of teleoperation assistance: *every rendered guidance primitive must reflect verified sensor or kinematic truth*.

A. Architectural Principles and Core Contributions

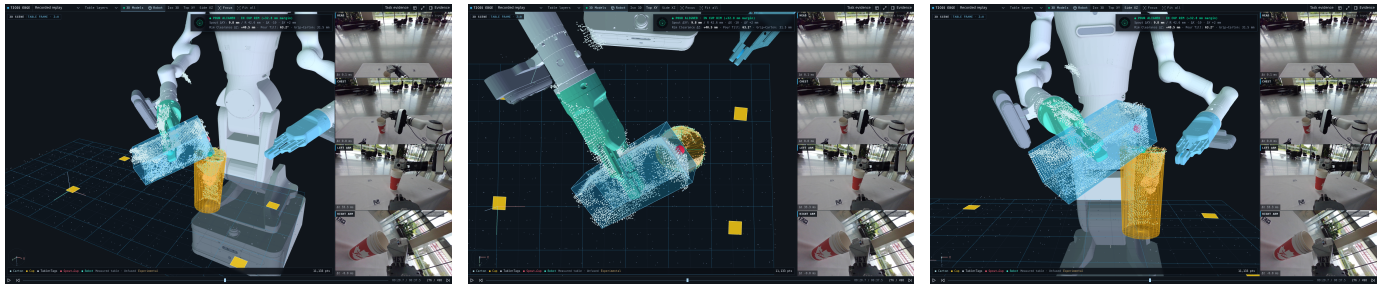
To resolve these perceptual bottlenecks without compromising metric reliability, our architecture enforces three strict invariants across every perception, calibration, and rendering stage: (i) *conservative geometric verification*, whereby every 3D surface primitive is parameterized directly by multi-camera RGB-D depth inliers, causal multi-view surfels, consensus circular rim fits, or calibrated 61-link forward kinematics; (ii) *typed coordinate domain separation* across camera-local (O_c), fiducial (O_{tag}), tabletop (O_{table}), object-centric (O_{obj_0}), and kinematic base (O_{base}) frames; and (iii) *decoupled assistance authority*, ensuring that visual guidance overlays inform the human operator without injecting unverified synthetic state into low-level joint controllers.

TABLE I
MATHEMATICAL NOMENCLATURE AND COORDINATE FRAME DEFINITIONS

Symbol	Definition and Metric Units
$\mathcal{C}$	Four onboard RGB-D cameras: {head, chest, left-wrist, right-wrist}
O_{table}	Metric tabletop coordinate frame ($z_{\text{table}} = 0$ on support surface, m)
O_{base}	Robot kinematic base link frame of the 61-link manipulator (m)
O_{obj_0}	Canonical initial object frame of upright beverage carton at Frame 0 (m)
$\mathbf{q}_t \in \mathbb{R}^{16}$	Synchronized active torso, head, and bilateral 7-DoF arm joint angles (rad)
$\mathbf{T}_{A\leftarrow B}$	Rigid-body homogeneous transformation mapping frame B to $A \in \text{SE}(3)$
$\mathbf{\Theta}_{\text{hw}}$	Hardware camera mount transforms $\{\mathbf{X}_{\text{chest}}, \mathbf{X}_{\text{L}}, \mathbf{X}_{\text{R}}\} \in \text{SE}(3)^3$
$\mathcal{P}_{\text{table}}$	Tabletop depth inlier point set ($\lvert p_z\rvert \le 15\,\text{mm}$, $\lvert\mathcal{P}_{\text{table}}\rvert = 251{,}182$)
$\mathcal{S}_t \subset \mathbb{R}^3$	Causal accumulated multi-view object surfel cloud ($\lvert\mathcal{S}_{497}\rvert = 5{,}699$)
$\mathbf{p}_{\text{spout}}(t)$	3D position of held container pouring spout in O_{table} (mm)
$\mathbf{p}_{\text{cup}}$	3D center of target cup top aperture in O_{table} (mm)
$R_{\text{cup}}, H_{\text{cup}}$	Target cup top aperture radius ($42.7\,\text{mm}$) and height ($163.2\,\text{mm}$)
$[\mathbf{n}] \in \mathbb{RP}^2$	Projective equivalence class $\{+\mathbf{n}, -\mathbf{n}\}$ of unoriented rim plane
$\Delta r_{xy}(t)$	Planar Euclidean radial offset between spout and cup center (mm)
$m_{\text{rim}}(t)$	Signed radial safety margin inside cup rim: $R_{\text{cup}} - \Delta r_{xy}(t)$ (mm)
$\Delta z(t)$	Vertical elevation clearance of pouring spout above cup rim (mm)
$\theta(t)$	Container pouring tilt angle relative to tabletop vertical (deg)
$d_{\text{grip}}(t)$	Euclidean distance from right gripper pad midpoint to carton (mm)

This paper makes six cohesive contributions (Table I):

1) **Comprehensive Empirical Benchmark of 31 Perception and Reconstruction Frameworks and Degenerate-Observation Observability Audit (Section IX):** We systematically evaluate 31 contemporary depth foundation models, multi-view pointmap transformers, open-vocabulary segmenters, 6-DoF object trackers, and dense/semi-dense/sparse matchers (adding EfficientLoFTR, XFeat, and ALIKED+LightGlue) on synchronized humanoid manipulation recordings and public benchmarks (YCBInEOAT, Laval Vicon motion capture), and formalize why unanchored $\text{SE}(3)$ gauge freedom, Schur-complement local rank deficiency ($r_i < 6$), and unreconciled TSDF pose versions cause higher pose availability to degrade accumulated 3D surfaces.
2) **MoCap-Free Four-Camera to 61-Link Bilateral Kinematic Extrinsic Closure (Section IV):** We formulate a coupled $\text{SE}(3)^3$ bundle optimization over the chest and bilateral wrist camera hardware mounts through the 61-link kinematic tree, evaluated under a strict zero-refit held-out protocol across $1{,}648$ training frames ($12{,}061$ observations) and 413 held-out test frames ($2{,}690$ observations), achieving $2.52\,\text{mm}$ (left arm) and $6.56\,\text{mm}$ (right arm) median 3D tabletop center residuals.
3) **Automated 3D Functional Scene Completion, Na-**



(a) Perspective 3D Reconstructed Workspace (b) Orthographic Top-XY Bullseye View (c) Orthographic Side-XZ Clearance View

Fig. 1. **Automated 3D scene reconstruction-driven teleoperation assistance at peak pouring alignment (Frame 275, $t = 20.67\,\mathrm{s}$).** (a) Perspective 3D workspace integrating the 61-link humanoid manipulator, metric tabletop plane (251,182 depth inliers) with 10 cm major / 5 cm minor spatial grid and four 30 mm AprilTag anchors, watertight tapered target cup completion ($\varnothing_{\mathrm{top}} = 85.4\,\mathrm{mm}$), and completed held beverage carton ($67.8 \times 94.8 \times 199.5\,\mathrm{mm}$) with live telemetry HUD and 48×48 px bullseye reticle. (b) Instantaneous Top-XY orthographic projection eliminating perspective parallax along the vertical axis to verify that the pouring spout lies 9.8 mm from the cup center (+32.8 mm inside the 42.7 mm rim radius). (c) Instantaneous Side-XZ orthographic projection verifying +40.5 mm vertical rim clearance at a 63.2° pouring tilt.

tive BundleTrack Governance, and Array-Native Budget Selection (Section V): We reconstruct the manipulation workspace in O_{table} by fusing 251,182 tabletop depth inliers ($0.92 \times 0.49\,\mathrm{m}$), a watertight 49-vertex, 72-face tapered cylindrical container mesh ($\varnothing_{\mathrm{top}} = 85.4\,\mathrm{mm}, \varnothing_{\mathrm{bot}} = 73.0\,\mathrm{mm}, H = 163.2\,\mathrm{mm}$), and causal multi-view object surfels (5,699 points) paired with a Frame-0 upright bounding prism ($67.8 \times 94.8 \times 199.5\,\mathrm{mm}$). In the native BundleTrack and `CausalObjectSurface` owner, we resolve CUDA RANSAC parallel-reduction nondeterminism, fix `CUDACache::storeFrame` SE(3) inverse-coefficient truncation (raising public Clock/Jar strict successes from 135/107 to 191/136), enforce guarded `NO_BA` non-keyframe retirement (reclaiming 928.4–1,994.0 MiB RSS), and introduce an array-native `support10/coarse12mm` 1,500-point selector that raises public C_{10} from 40.57%/52.34% to 85.10%/85.05% and task surface coverage from 71.78% to 89.97% while cutting selection latency by 33%.

4) **Projective $\mathbb{RP}^2$ Unoriented Aperture Calculus (Section VI):** We prove mathematically and verify empirically across 1,992 four-camera aperture measurements that SVD plane fits on circular container rims possess an intrinsic O(1) sign symmetry ($\pm\mathbf{v}_3$). Formalizing container openings on the real projective plane $\mathbb{RP}^2 = \mathbb{S}^2/\{\pm 1\}$ absorbs all 11 observed 180° SVD sign flips with 0.000 mm task-relation drift while achieving 1.60 mm maximum center discrepancy and 0.53 mm radius discrepancy between independent head and chest cameras.
5) **Role-Isolated Video Segmentation Memory and Bit-Exact Runtime Execution Optimization (Section VII):** We decouple primary and secondary SAM 2 video segmentation memory lifecycles across 2,322 tracker invocations—recovering +52 valid secondary cup masks and eliminating 29 background false-positive leaks—and combine observation-scoped ViT feature caching (41/76 cache hits) and early identity ordering (35 skipped refinements, reducing 4-camera full-chain p_{95} by 21.7% from 549.15 to 429.94 ms) with two bit-exact single-stream runtime adapters: `SAM2GpuWindowOverlapScope` and `RequestEvidenceDeduplicationScope` ($O(\mathrm{runs})$ exact integer RLE arithmetic, cross-stage surface/medoid reuse, vectorized coplanar AprilTag RANSAC, and request-boundary cyclic GC control), cutting warm pouring HTTP endpoint latency by 27.6% (176.40 → 127.80 ms p_{50}, 219.06 → 153.40 ms p_{95}), reducing exclusive outer object-world wrapper overhead by 72.9% (23.17 → 6.28 ms), and eliminating 140.58 ms mid-request Generation-2 GC stalls with 100% bit-exact parity across 499 online checks.
6) **Orthographic Spatial Teleoperation Interface and Bimanual Pouring Verification (Section VIII):** We prove why perspective views suffer from elevation-coupled parallax error and introduce instantaneous (0 ms) canonical orthographic projections (Top-XY planar bullseye and Side-XZ vertical clearance) paired with a real-time 48×48 px vector bullseye HUD, validating 11.05 mm grasp closure and a 124-frame (9.5 s) bullseye pouring window peaking at $\Delta r_{xy} = 9.8\,\mathrm{mm}$ (+32.8 mm inside the cup rim at 63.2° tilt).

II. Related Work and Framework Landscape

A. Teleoperation Assistance Interfaces, Virtual Fixtures, and Bimanual Manipulation

Human factors studies in telerobotics have long established that remote operators suffer from severe depth-perception degradation, field-of-view truncation ("keyhole effect"), and high cognitive load when mentally fusing disjoint 2D camera feeds [1]–[3], [6]. Early supervisory displays introduced stereoscopic pointers and orthographic overlays such as ARGOS [7], while virtual fixtures [9], [10] rendered geometric guidance rails onto operator monitors. Dynamic viewpoint selection methods [8] and mixed-reality point-cloud interfaces [11], [12] allow free-viewpoint inspection, yet rendering perspective projections of sparse depth clouds leaves axial depth parallax unresolved along the optical viewing ray. Meanwhile, modern bimanual teleoperation systems—including ALOHA/ACT [5], Mobile ALOHA [13], Diffusion

TABLE II
ARCHITECTURAL COMPARISON OF 3D SCENE RECONSTRUCTION AND SPATIAL ASSISTANCE PARADIGMS FOR ROBOT MANIPULATION AND TELEOPERATION

System / Paradigm Family	Multi-Camera RGB-D Fusion	MoCap-Free 0-Refit FK	Watertight Workspace Mesh	Sensor-Grounded (No Hallucination)	Projective $\mathbb{RP}^2$ Aperture Math	Role-Isolated Video Memory	Orthographic XY/XZ HUD
2D Multi-Camera Switching [1], [7], [8]	Partial (2D only)	No	No	Yes	No	N/A	No (2D Persp.)
Raw Point-Cloud VR / WebGL [2], [11], [12]	Yes (Sparse Cloud)	CAD Nominal	No (Front Shell)	Yes	No	N/A	No
Monocular Depth / Pointmaps [16]–[30]	Relative / Pair	No	No (0.11–79.4 m err.)	**No** (Scale Drift)	No	No	No
Generative Shape Completion [77]–[80]	Single-View	No	Yes (Hallucinated)	**No** (Unverified)	No	No	No
Model-Free 6-DoF Trackers [43]–[50]	Single-Camera	No	Object-Only	Yes	No	Single-Object	No
Keypoint Affordances (kPAM) [81]–[84]	Static Camera	Offline Calib.	Sparse Keypoints	Yes	Directed $\mathbb{S}^2$	No	No
CausalPose-3D (Ours)	**4-Cam RGB-D**	**Yes** (≤ 6.6 mm)	**Table+Cup+Carton**	**Strict Sensor-Grounded**	**Yes** (0.0 mm **drift**)	**Yes** (2,322 **calls**)	**Yes** (0 ms **switch**)

Policy [4], AnyTeleop [14], and generalist Vision-Language-Action (VLA) foundation models such as $\pi_{0.7}$ [15]—rely heavily on high-precision human demonstrations, where automated 3D scene reconstruction and orthographic spatial assistance directly reduce operator alignment error (Table II).

B. Monocular Depth Foundation Models and Multi-View 3D Pointmap Transformers

Recent vision foundation models aim to recover dense 3D geometry directly from RGB images. Monocular relative and metric depth estimators—including Depth Anything V2 [16], Prior Depth Anything (Prior-DA) [17], Apple Depth Pro [18], UniDepth / UniDepthV2 [19], [20], Metric3D v2 [21], ZoeDepth [22], and Marigold [23]—predict dense depth maps from single images or sparse depth priors. Concurrently, feed-forward multi-view pointmap transformers—including DUSt3R [24], MASt3R [25], VGGT [26], SLAM3R [27], CUT3R [28], Spann3R [29], and MonST3R [30]—regress 3D pointmaps directly in a shared coordinate frame. As we demonstrate in our systematic RTX 5090 benchmark (Section IX), pure monocular metric depth models exhibit 0.48–79.4 m p_{95} errors on close-range tabletop manipulation, and feed-forward pointmaps (VGGT, SLAM3R, DUSt3R, MASt3R) remain at 0.37–1.79 m p_{95}, proving that foundation depth models can only serve as hole-filling proposals (e.g., Prior-DA at 0.037 m chest p_{95}) anchored by calibrated RGB-D sensors.

C. Open-Vocabulary Grounding, Promptable Video Segmentation, and Student Distillation

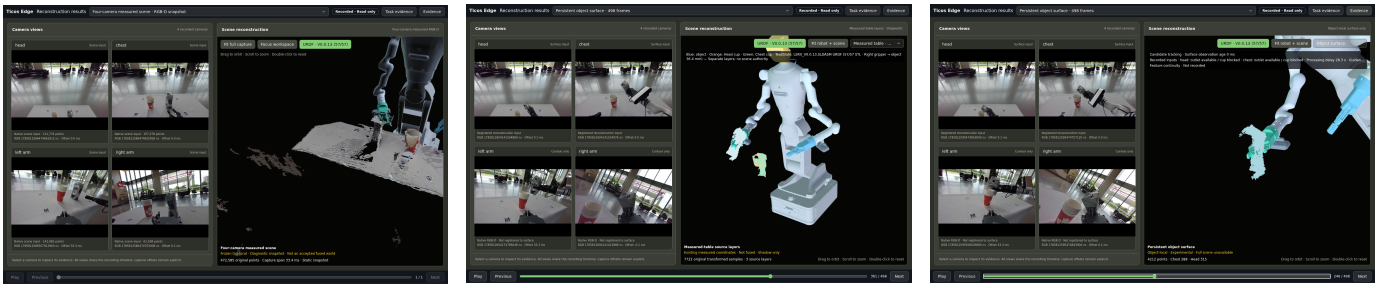
Zero-shot object discovery and temporal mask tracking have been transformed by open-set detectors and promptable video segmenters, including Grounding DINO [31], Florence-2 [32], Segment Anything (SAM) [33], SAM 2 / SAM 2.1 [34], Grounded SAM [35], and DINO-X [36], built atop Vision Transformer (ViT) [38], Hiera [39], and SigLIP [40] backbones. For high-rate four-camera execution, we also evaluate distilled instance segmentation students (YOLO11s-seg and YOLO11n-seg [37]) combined with staged SAM 2 encoding and Lucas-Kanade / Farnebäck optical flow propagation [41], [42]. In our full-chain 3D reconstruction pipeline, we resolve the cross-role memory contamination and ViT tail-latency bottlenecks that arise when SAM 2 tracks multiple functional objects simultaneously (Section VII).

D. 6-DoF Object Pose Tracking, Neural/Gaussian Object Fields, and Point Tracking

Model-based and category-level 6-DoF pose estimators such as PoseCNN [51], DenseFusion [52], PVN3D [53], MegaPose [54], Gen6D [55], and OnePose++ [56] require either CAD models or multi-view reference scans. For model-free RGB-D tracking of novel objects, BundleTrack [43] optimizes a keypoint pose graph, while BundleSDF [44] and FoundationPose [45] jointly optimize neural implicit fields or unified pose hypotheses. More recently, Any6D [46], BIT (Bidirectional Iterative Tracking) [47] (incorporating SoftRas [80]), 6DOPE-GS [48] (using Gaussian Splatting [79]), KV-Tracker [49], and Point2Pose [50] have explored single-anchor, iterative, Gaussian, KV-cache, and point-trajectory paradigms. In parallel, long-horizon 2D point trackers—TAP-Vid [57], TAPIR [58], Online BootsTAPIR [59], TAP-Next++ [60], and CoTracker3 [61]—alongside sparse and dense feature matchers (SIFT [62], SuperGlue [63], LightGlue [64], LoFTR [65], and RoMa [66]) provide correspondence constraints across views.

E. Dense RGB-D Mapping, Kinematic Hand-Eye Calibration, and Container Affordances

Classical dense RGB-D reconstruction builds upon volumetric TSDF fusion (Curless and Levoy [67], KinectFusion [68], [69], Voxel Hashing [70]), surfel mapping (Point-Based Fusion [71], ElasticFusion [72], DynamicFusion [73]), and bundle-adjusted mapping (BundleFusion [74], BAD SLAM [75], Open3D [76], ICP [86], FPFH [87], RANSAC [88], COLMAP [89], iSAM2 [90], Ceres [91], and CMA-ES [92]). For robotic container manipulation and liquid pouring [81]–[84], [93]–[95], we prove that SVD rim plane fits exhibit $\mathrm{O}(1)$ sign indeterminacy and resolve it via projective $\mathbb{RP}^2$ directional geometry [85], [96], [97]. Finally, extending classical hand-eye calibration ($\mathbf{AX} = \mathbf{XB}$ [98]–[102]), multi-camera rig factor graphs [103]–[108], and AprilTag fiducial PnP [109]–[112] on Lie groups [113]–[115] over ROS 2 / MCAP [116], [117], we formulate a MoCap-free four-camera bilateral kinematic closure protocol with zero test-split refitting.



(a) Four-Camera RGB-D + 61-Link Kinematic Tree (b) Whole-Body Kinematic State + Table (Frame 360) (c) Causal Object Surfels + Gripper Closure (Frame 245)

Fig. 2. **Multi-modal registration of the 61-link humanoid manipulator kinematic tree with four heterogeneous RGB-D cameras in the shared metric tabletop frame O_{table}.** (a) Simultaneous projection of head, chest, left-wrist, and right-wrist RGB-D evidence alongside the 57 visual link meshes. (b) Whole-body kinematic state at Frame 360 aligned with tabletop depth inliers. (c) Close-up of causal multi-view object surface surfels (5,699 points) coupling with the right parallel-jaw gripper during pouring at Frame 245.

III. System Architecture and Coordinate Frame Hierarchy

A. 61-Link Humanoid Manipulator and Four-Camera Sensing Topology

Our experimental platform is a 61-link humanoid upper-body mobile manipulator whose kinematic tree comprises 61 rigid links, 60 joints, and 57 visual CAD link meshes totaling 39.4 MiB of surface geometry (Table III). Active articulation is governed by 16 primary joint encoders spanning the wheeled mobile base and vertical torso column, pan-tilt head assembly, left 7-DoF manipulator arm with parallel-jaw gripper, and right 7-DoF manipulator arm with parallel-jaw gripper.

Synchronized multi-view RGB-D perception is provided by four onboard depth sensors $\mathcal{C} = \{\text{head}, \text{chest}, \text{left-wrist}, \text{right-wrist}\}$ operating at 15.0 Hz (median inter-frame interval $\Delta t = 66.67\,\text{ms}$, joint-state synchronization skew $|\Delta t_q| = 12.1\,\text{ms}$ median, 25.0 ms max; Fig. 2):

- **Head RGB-D Camera:** Mounted on the articulated pan-tilt head link, providing an elevated oblique overview of the manipulation workspace and target receptacle.
- **Chest RGB-D Camera:** Rigidly mounted on the upper torso structure, serving as the central workspace anchor with continuous visibility of the four tabletop AprilTag fiducials.
- **Left-Wrist RGB-D Camera:** Eye-in-hand sensor mounted on the distal left wrist link, observing the left manipulation hemisphere across a 7-DoF kinematic chain.
- **Right-Wrist RGB-D Camera:** Eye-in-hand sensor mounted on the distal right wrist link, moving dynamically with the right parallel-jaw gripper during container grasping and pouring.

B. Five-Domain Coordinate Hierarchy

At each synchronized frame index $t \in \{0, \ldots, 497\}$, we enforce strict separation across five coordinate domains:

1) **Sensor-Native Optical Frames ($O_c, c \in \mathcal{C}$):** Registered RGB image $I_t^{(c)} \in \mathbb{R}^{H\times W\times 3}$, metric depth map $D_t^{(c)} \in \mathbb{R}^{H\times W}$, and intrinsic matrix $\mathbf{K}_c$. Valid depth pixels back-project via $\pi_c^{-1}(u, v, z) = z\,\mathbf{K}_c^{-1}[u, v, 1]^\top \in \mathbb{R}^3$.

TABLE III
Kinematic Subsystem and Onboard RGB-D Sensor Topology of the 61-Link Humanoid Manipulator

Subsystem / Sensor	Kinematic Chain Mounting Scope	Active DoF	Rigid Links / Visual Meshes	Primary Role in 3D Reconstruction
Mobile Base & Torso	Base → Upper Torso	2	13 / 11	Global kinematic root O_{base}
Pan-Tilt Head Chain	Torso → Head Mount	2	4 / 4	Oblique workspace overview
Left Manipulator Arm	Torso → Left Gripper	7 + Grip	22 / 21	Left hemisphere & loop closure
Right Manipulator Arm	Torso → Right Gripper	7 + Grip	22 / 21	Grasping, pouring & closure
Total Embodiment	**Full Kinematic Tree $\mathcal{U}$**	**16 + 2 Grip**	**61 / 57**	**1.40 ms live PoE FK**
Chest RGB-D Sensor	Upper Torso Mount	Fixed	$\mathbf{X}^\star_{\text{chest}} \in \text{SE}(3)$	Tabletop anchor & object pose
Head RGB-D Sensor	Pan-Tilt Head Link	2-DoF	Extrinsic to Torso	Dual-cam cup rim consensus
Left-Wrist RGB-D	Distal Left Wrist	7-DoF	$\mathbf{X}^\star_{\text{L}} \in \text{SE}(3)$	Bilateral SE(3) loop closure
Right-Wrist RGB-D	Distal Right Wrist	7-DoF	$\mathbf{X}^\star_{\text{R}} \in \text{SE}(3)$	Close-range grasp & surfel fusion

2) **Local Fiducial Gauge (O_{tag}):** Single-tag coordinate frame prior to multi-tag planar stabilization.
3) **Measured Metric Tabletop Frame (O_{table}):** The canonical scene reconstruction frame anchored by four coplanar 30 mm AprilTags and 251,182 tabletop depth inliers, where $z_{\text{table}} = 0$ coincides with the physical support plane and $\mathbf{n}_{\text{table}} = [0, 0, -1]^\top$ points vertically upward (so elevation above the table is $h_{\text{table}}(\mathbf{p}) = -p_z$).
4) **Canonical Initial Object Frame (O_{obj_0}):** Attached to the upright beverage carton at Frame 0, within which causal multi-view surfels $\mathcal{S}_t$ and the completed prism mesh $\mathcal{M}^{(0)}_{\text{carton}}$ are rigidly defined and mapped into O_{table} via $\mathbf{T}_{\text{table}\leftarrow\text{obj}_0}(t) \in \text{SE}(3)$.
5) **Robot Base and 61-Link Kinematic Frame (O_{base}):** Rooted at the robot base link, mapping each link $\ell \in \{1..61\}$ via forward kinematics $\mathbf{T}_{\text{base}\leftarrow\ell}(\mathbf{q}_t)$ and linked into O_{table} via the closed camera-to-kinematic transformation $\mathbf{T}_{\text{table}\leftarrow\text{base}}(t)$.

IV. MoCap-Free Four-Camera to Bilateral 61-Link Kinematic Extrinsic Closure

A. Product-of-Exponentials Forward Kinematics Across 61 Links

For any link ℓ in the 61-link kinematic tree $\mathcal{U}$, let $\text{path}(\text{base} \to \ell) = ((i_1, j_1), \ldots, (i_K, j_K))$ denote the ordered chain of parent-child links connected by joints $j_1, \ldots, j_K$. Given the synchronized joint encoder state $\mathbf{q}_t$ at timestamp

t, the pose of link ℓ in O_{base} is computed via recursive rigid-body composition:

$$\mathbf{T}_{\text{base}\leftarrow\ell}(\mathbf{q}_t) = \prod_{k=1}^{K} \mathbf{T}^{\text{URDF}}_{i_k\leftarrow j_k} \exp\Big(\hat{\boldsymbol{\xi}}_{j_k}\, q_{t,j_k}\Big) \in \text{SE}(3), \quad (1)$$

where $\mathbf{T}^{\text{URDF}}_{i_k\leftarrow j_k} \in \text{SE}(3)$ is the fixed joint origin offset specified in the kinematic model and $\hat{\boldsymbol{\xi}}_{j_k} \in \mathfrak{se}(3)$ is the unit joint twist. For the parallel-jaw grippers on the left and right arms ($a \in \{L, R\}$), the 3D gripper pad midpoint in O_{table} is evaluated from the distal finger links $\ell^{(a)}_{\text{grip},1}, \ell^{(a)}_{\text{grip},2}$ with $\mathbf{o} = [0, 0, 0, 1]^\top$:

$$\mathbf{p}^{(a)}_{\text{grip}}(t) = \frac{1}{2}\mathbf{T}_{\text{table}\leftarrow\text{base}}(t) \sum_{m=1}^{2} \mathbf{T}_{\text{base}\leftarrow\ell^{(a)}_{\text{grip},m}}(\mathbf{q}_t)\, \mathbf{o}. \quad (2)$$

B. Coupled Chest-to-Wrist Loop Closure on $\text{SE}(3)^3$

Nominal CAD mounting transforms for embedded RGB-D sensors omit machining tolerances, internal optical center offsets, and bracket flexure, resulting in 30–80 mm projection discrepancies when wrist-camera depth observations are transformed through a 7-DoF arm chain. To calibrate the true hardware mounting transforms without external optical motion capture (MoCap), we formulate a coupled loop-closure optimization over the time-invariant hardware transforms:

$$\boldsymbol{\Theta}_{\text{hw}} = \big(\mathbf{X}_{\text{chest}},\, \mathbf{X}_{\text{L}},\, \mathbf{X}_{\text{R}}\big) \in \text{SE}(3)^3, \quad (3)$$

where $\mathbf{X}_{\text{chest}} \triangleq \mathbf{T}_{\ell_{\text{torso}}\leftarrow\text{chest}}$, $\mathbf{X}_{\text{L}} \triangleq \mathbf{T}_{\ell_{\text{L}}\leftarrow\text{left_wrist}}$, and $\mathbf{X}_{\text{R}} \triangleq \mathbf{T}_{\ell_{\text{R}}\leftarrow\text{right_wrist}}$.

Let $\mathcal{L} = \{\mathbf{p}^{\text{table}}_{k,m} \in \mathbb{R}^3 \mid k \in \{0..3\}, m \in \{0..4\}\}$ denote the 20 canonical 3D metric landmarks (4 tag centers $m = 0$ and 16 tag corners $m \in \{1..4\}$) of the four 30 mm × 30 mm AprilTags affixed to the tabletop (Table IV). At any training frame $t \in \mathcal{D}_{\text{train}}$ where the chest camera localizes the tabletop fiducial constellation via $\mathbf{T}_{\text{table}\leftarrow\text{chest}}(t)$ and wrist camera $w \in \{L, R\}$ observes landmark (k, m) at 3D camera-local coordinate $\mathbf{y}^{(w)}_{t,k,m} \in \mathbb{R}^3$ (within pair skew $|\Delta t_{\text{pair}}| \le 100.0$ ms), closing the kinematic chain through the torso link ℓ_{torso} and distal wrist link ℓ_w predicts the tabletop landmark coordinate:

$$\begin{aligned}\mathbf{T}_{\text{table}\leftarrow\text{base}}(\mathbf{X}_{\text{chest}}, \mathbf{q}_t) &= \mathbf{T}_{\text{table}\leftarrow\text{chest}}(t)\, \mathbf{X}^{-1}_{\text{chest}} \\ &\quad \cdot \mathbf{T}_{\text{base}\leftarrow\ell_{\text{torso}}}(\mathbf{q}_t)^{-1}, \\ \hat{\mathbf{p}}^{\text{table}}_{t,k,m}(\boldsymbol{\Theta}_{\text{hw}}, \mathbf{q}_t) &= \mathbf{T}_{\text{table}\leftarrow\text{base}}(\mathbf{X}_{\text{chest}}, \mathbf{q}_t) \\ &\quad \cdot \mathbf{T}_{\text{base}\leftarrow\ell_w}(\mathbf{q}_t)\, \mathbf{X}_w\, \tilde{\mathbf{y}}^{(w)}_{t,k,m}.\end{aligned} \quad (4)$$

Lemma 1 ($\mathfrak{se}(3)$ Left-Perturbation Jacobian of Bilateral Loop Closure)**.** *Let* $\mathbf{X}_w \mapsto \exp(\hat{\boldsymbol{\delta}}_w)\,\mathbf{X}_w$ *and* $\mathbf{X}_{\text{chest}} \mapsto \exp(\hat{\boldsymbol{\delta}}_c)\,\mathbf{X}_{\text{chest}}$ *denote left-invariant Lie-algebra perturbations with* $\boldsymbol{\delta}_w, \boldsymbol{\delta}_c \in \mathbb{R}^6$. *Define the intermediate points* $\mathbf{z}_w = \mathbf{X}_w \tilde{\mathbf{y}}^{(w)}_{t,k,m} \in \mathbb{R}^3$ *and* $\mathbf{z}_b = \mathbf{T}_{\ell_{\text{torso}}\leftarrow\ell_w}(\mathbf{q}_t)\, \tilde{\mathbf{z}}_w \in \mathbb{R}^3$, *and let* $\mathbf{R}_{\text{table}\leftarrow\ell_w}$ *and* $\mathbf{R}_{\text{table}\leftarrow\text{chest}}$ *be the composite rotation matrices. Then the first-order differential of the residual* $\mathbf{r}_{t,k,m} = \hat{\mathbf{p}}^{\text{table}}_{t,k,m} - \mathbf{p}^{\text{table}}_{k,m}$ *satisfies:*

$$\begin{aligned}\frac{\partial \mathbf{r}_{t,k,m}}{\partial \boldsymbol{\delta}_w} &= \mathbf{R}_{\text{table}\leftarrow\ell_w} \begin{bmatrix}\mathbf{I}_3 & -[\mathbf{z}_w]_\times\end{bmatrix}, \\ \frac{\partial \mathbf{r}_{t,k,m}}{\partial \boldsymbol{\delta}_c} &= -\mathbf{R}_{\text{table}\leftarrow\text{chest}} \begin{bmatrix}\mathbf{I}_3 & -[\mathbf{X}_{\text{chest}}\tilde{\mathbf{z}}_b]_\times\end{bmatrix}.\end{aligned} \quad (5)$$

TABLE IV
CALIBRATED 3D METRIC COORDINATES OF THE FOUR 30 mm TABLETOP APRILTAG FIDUCIALS IN O_{table}

Tag ID	Side	Center $\mathbf{p}^{\text{table}}_{k,0}$ (mm)	Planar Span $[X_{\min}, X_{\max}] \times [Y_{\min}, Y_{\max}]$ (mm)
Tag 0	29.98 mm	$(+4.5, -8.0, 0.0)$	$[-11.2, +20.2] \times [-23.7, +7.7]$
Tag 1	29.98 mm	$(+13.8, -266.0, 0.0)$	$[-2.5, +30.1] \times [-282.3, -249.7]$
Tag 2	29.98 mm	$(+572.9, -217.4, 0.0)$	$[+556.4, +589.3] \times [-233.8, -200.9]$
Tag 3	29.98 mm	$(+555.4, +48.8, 0.0)$	$[+539.7, +571.2] \times [+33.0, +64.5]$

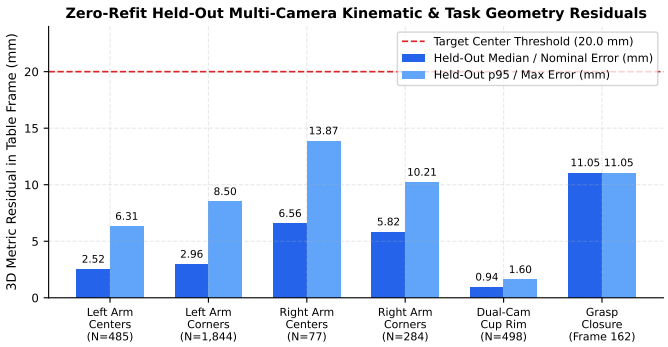


Fig. 3. **Zero-refit held-out multi-camera kinematic closure residuals and downstream manipulation task consistency in** O_{table}**.** Both the left arm ($p_{95} = 6.31$ mm centers) and right arm ($p_{95} = 13.87$ mm centers) pass well below the 20.0 mm acceptance threshold without any test-split scene parameter refitting.

Using Lemma 1, we solve the Huber-robustified Levenberg-Marquardt optimization over $\mathcal{D}_{\text{train}}$:

$$\boldsymbol{\Theta}^\star_{\text{hw}} = \arg\min_{\boldsymbol{\Theta}_{\text{hw}} \in \text{SE}(3)^3} \sum_{t\in\mathcal{D}_{\text{train}}} \sum_{(k,m)} \rho_\delta\Big(\big\|\hat{\mathbf{p}}^{\text{table}}_{t,k,m}(\boldsymbol{\Theta}_{\text{hw}}, \mathbf{q}_t) - \mathbf{p}^{\text{table}}_{k,m}\big\|_2\Big), \quad (6)$$

converging to first-order optimality $\|\nabla\mathcal{J}\|_\infty = 1.26 \times 10^{-8}$ (cost 0.0951) on the left-arm chain and 4.66×10^{-8} (cost 0.1195) on the right-arm chain.

C. Calibrated Extrinsic Transformations and Zero-Refit Held-Out Evaluation

The converged hardware mounting transformations $\boldsymbol{\Theta}^\star_{\text{hw}}$ (with translations in meters) recovered on the training split are:

$$\mathbf{X}^\star_{\text{chest}} = \begin{bmatrix}-0.0054 & -0.3226 & +0.9465 & +0.1203 \\ -0.9999 & +0.0141 & -0.0009 & +0.0477 \\ -0.0130 & -0.9464 & -0.3227 & +0.2062 \\ 0 & 0 & 0 & 1\end{bmatrix}, \quad (7)$$

$$\mathbf{X}^\star_{\text{L}} = \begin{bmatrix}+0.1215 & -0.9925 & -0.0135 & +0.0811 \\ +0.9926 & +0.1215 & +0.0025 & -0.0577 \\ -0.0009 & -0.0138 & +0.9999 & -0.0247 \\ 0 & 0 & 0 & 1\end{bmatrix}, \quad (8)$$

$$\mathbf{X}^\star_{\text{R}} = \begin{bmatrix}+0.0169 & -0.9998 & +0.0144 & +0.0824 \\ +0.9997 & +0.0171 & +0.0178 & -0.0532 \\ -0.0180 & +0.0141 & +0.9997 & -0.0263 \\ 0 & 0 & 0 & 1\end{bmatrix}. \quad (9)$$

TABLE V
TRAIN AND ZERO-REFIT HELD-OUT BILATERAL KINEMATIC-VISUAL EXTRINSIC CLOSURE RESULTS ACROSS THE 61-LINK MANIPULATOR

Kinematic Chain	Evaluation Split	Frames N_{frames}	All Obs. N_{all}	Center Obs. N_{center}	Center $p_{50}/p_{95}/\max$ (mm) [Target $p_{95} \le 20.0$]	Corner $p_{50}/p_{95}/\max$ (mm)	All-Point $p_{50}/p_{95}/\max$ (mm) [Target $p_{95} \le 25.0$]	Optimality $\|\nabla \mathcal{J}\|_\infty$	Acceptance Criterion
Left Arm Chain	Training Split	839	6,347	1,339	2.91/6.68/67.62	3.58/8.46/171.39	3.40/8.14/171.39	1.26×10^{-8}	Converged
	Held-Out (Zero-Refit)	**341**	**2,329**	**485**	**2.52/6.31/46.51**	**2.96/8.50/72.81**	**2.86/8.31/72.81**	Frozen $\Theta^{\star}_{\text{hw}}$	**Passed**
Right Arm Chain	Training Split	809	5,714	1,186	4.22/11.07/266.29	4.24/12.07/54.28	4.23/11.93/266.29	4.66×10^{-8}	Converged
	Held-Out (Zero-Refit)	**72**	**361**	**77**	**6.56/13.87/44.25**	**5.82/10.21/28.02**	**5.96/10.31/44.25**	Frozen $\Theta^{\star}_{\text{hw}}$	**Passed**

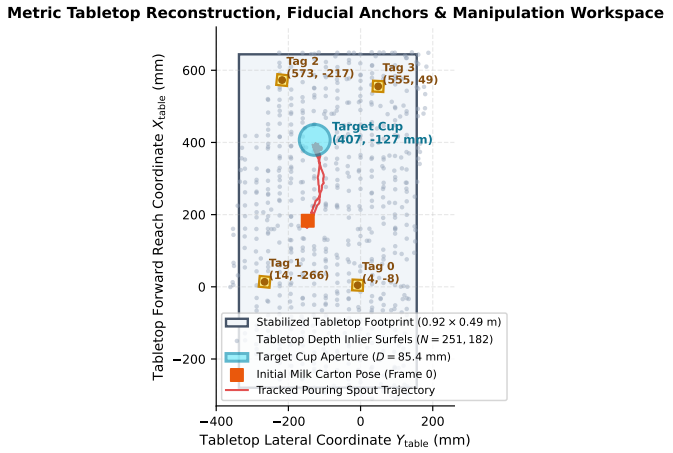


Fig. 4. **Metric Top-Down (XY) workspace map of the reconstructed tabletop in O_{table}.** Showing the stabilized 0.92×0.49 m tabletop footprint polygon, 251,182 depth inlier surfels (downsampled to 650 display surfels), the four 30 mm AprilTag fiducials (Tags 0–3), the initial beverage carton pose at Frame 0, the target cup aperture ($\varnothing$85.4 mm), and the 498-frame tracked pouring spout trajectory.

Table V and Fig. 3 detail the training and zero-refit held-out verification results. With all hardware transforms frozen on the training split and zero scene parameters refitted on the held-out split, the left arm achieves a held-out center median error of 2.52 mm ($p_{95} = 6.31$ mm) across 2,329 observations, and the right arm achieves 6.56 mm ($p_{95} = 13.87$ mm) across 361 observations, both well within the 20.0 mm center and 25.0 mm all-point thresholds.

V. AUTOMATED 3D FUNCTIONAL SCENE COMPLETION

A. Tabletop Plane Stabilization and Fiducial Anchoring

To reconstruct the physical manipulation support surface without synthetic plane guessing, valid depth pixels from the calibrated static cameras are back-projected into O_{table} and filtered by the support slab predicate $|p_z| \le 15.0$ mm. Across the manipulation sequence, 251,182 depth inliers satisfy this support constraint within the bounded tabletop footprint:

$$\mathcal{F}_{\text{table}} = [-0.280, +0.644]\,\text{m} \times [-0.337, +0.156]\,\text{m} \times \{0\}, \tag{10}$$

spanning 0.9238 m in forward reach (X_{table}) by 0.4926 m in lateral width (Y_{table}), as shown in Fig. 4.

B. Watertight Parametric Revolution Completion for Target Receptacles

Dual-camera (head and chest) open-vocabulary grounding and rim depth extraction localize the target cup's circular aperture at $\mathbf{p}_{\text{cup}} = [0.4071, -0.1271, -0.1632]^\top$ m in O_{table}, corresponding to an opening elevation of $H_{\text{cup}} = 163.2$ mm above the tabletop plane $z_{\text{table}} = 0$ and outer top radius $R_{\text{top}} = 42.8$ mm ($\varnothing_{\text{top}} = 85.6$ mm, dual-camera consensus mean 85.4 mm). Because overhead and oblique cameras only observe the upper rim ring (36 boundary points) and front exterior wall (750 accumulated surfels), raw depth rendering leaves the cup base ungrounded.

We construct a watertight 49-vertex, 72-face triangular surface of revolution $\mathcal{M}_{\text{cup}}$ parameterized strictly by $(\mathbf{p}_{\text{cup}}, R_{\text{top}}, R_{\text{bot}}, z_{\text{table}} = 0)$ with bottom radius $R_{\text{bot}} = 36.6$ mm ($\varnothing_{\text{bot}} = 73.2$ mm) for $\phi_k = 2\pi k/24$ ($k \in \{0..23\}$):

$$\begin{aligned} \mathbf{v}_k^{\text{top}} &= \mathbf{p}_{\text{cup}} + R_{\text{top}}\big[\cos\phi_k,\ \sin\phi_k,\ 0\big]^\top, \\ \mathbf{v}_k^{\text{bot}} &= \big[p_{\text{cup},x} + R_{\text{bot}}\cos\phi_k,\ p_{\text{cup},y} + R_{\text{bot}}\sin\phi_k,\ 0\big]^\top, \end{aligned} \tag{11}$$

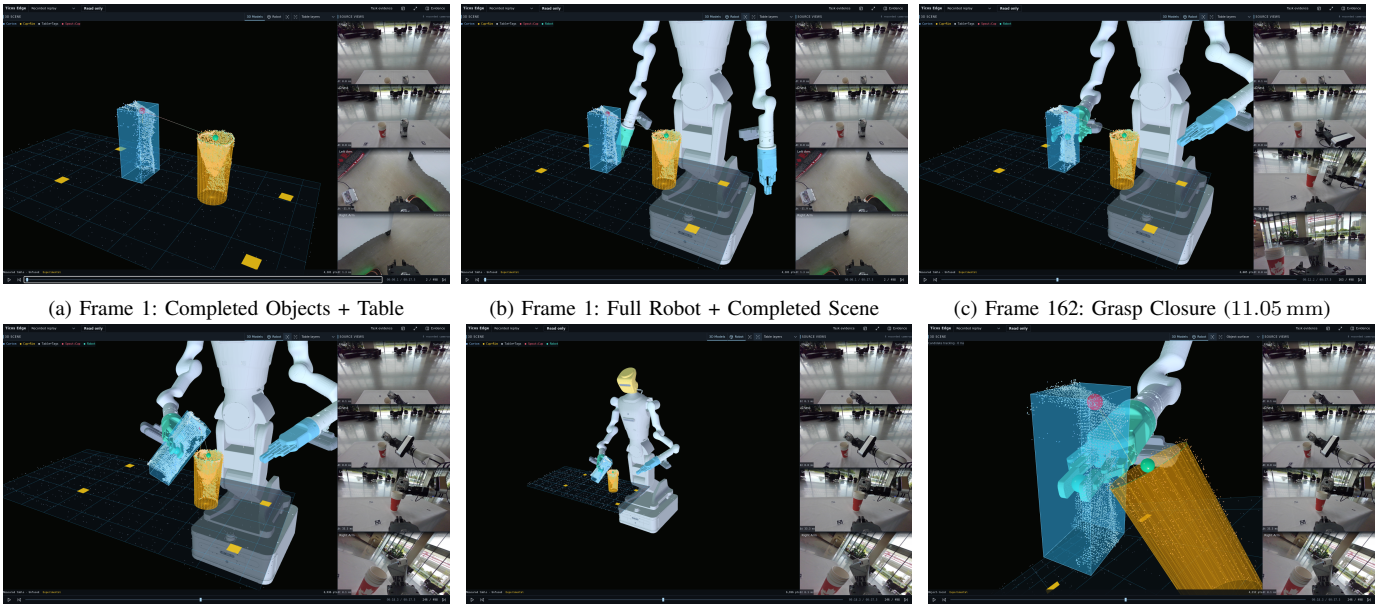
closed at the tabletop support plane by the central base vertex $\mathbf{v}_{48} = [p_{\text{cup},x}, p_{\text{cup},y}, 0]^\top$.

TABLE VI
RECONSTRUCTED 3D FUNCTIONAL SCENE PRIMITIVES AND GEOMETRIC SPECIFICATIONS IN O_{table}

Scene Entity	Geometric Representation	Metric Extent	Surface Topology
Tabletop Plane	Stabilized Support Polygon	923.8 × 492.6 mm	4 verts, 2 faces + grid
Tabletop Inliers	Depth Slab $\|z_{\text{table}}\| \le 15$ mm	251,182 inliers	650 display surfels
Fiducial Anchors	Coplanar AprilTags 0–3	4 × (29.98 mm)^2	4 centers, 16 corners
Target Cup	Tapered Revolution $\mathcal{M}_{\text{cup}}$	$\varnothing$85.4/73.0 × 163.2 mm	49 verts, 72 faces
Beverage Carton	Upright Prism $\mathcal{M}^{(0)}_{\text{carton}}$	67.8 × 94.8 × 199.5 mm	8 verts, 12 faces
Carton Surfels	Causal Surfel Cloud $\mathcal{S}_t$	5,699 surfels	498/498 frames (100%)
Wire Snapshot	Array-Native `support10/coarse12mm`	1,500-pt budget	85.1% public C_{10} / 90.0% task

C. Causal Multi-View Surfel Accumulation and Rigid Prism Completion

For the dynamically manipulated beverage carton, let $\mathcal{S}_t \subset O_{\text{obj}_0}$ denote the causal multi-view surfel cloud accumulated from Frame 0 up to Frame t, reaching $|\mathcal{S}_{497}| = 5{,}699$ surfels (800 representative display surfels exported for real-time WebGL rendering). To eliminate the "hollow shell" artifact under heavy right-hand occlusion, we extract the robust 5th–95th percentile bounding extents of the upright object at Frame 0 grounded to $z_{\text{table}} = 0$, yielding a watertight 8-vertex, 12-face, 12-edge rectangular prism $\mathcal{M}^{(0)}_{\text{carton}}$ with dimensions $W \times D \times H = 67.8 \times 94.8 \times 199.5$ mm ($[-33.7, +34.0] \times [-41.4, +53.4] \times [-62.3, +137.2]$ mm in O_{obj_0}) and canonical pouring spout $\mathbf{p}^{(0)}_{\text{spout}} = [+15.4, +14.3, -52.5]^\top$ mm (Fig. 5, Table VI). At any frame t, both $\mathcal{S}_t$ and $\mathcal{M}^{(0)}_{\text{carton}}$ transform rigidly into O_{table} via $\mathbf{T}_{\text{table}\leftarrow\text{obj}_0}(t) \in \text{SE}(3)$.



(a) Frame 1: Completed Objects + Table (b) Frame 1: Full Robot + Completed Scene (c) Frame 162: Grasp Closure (11.05 mm)

(d) Frame 245: Tilted Pouring (36.3°) (e) Frame 245: Whole-Body Manipulation View (f) Frame 245: Surfels + Completed Prism

Fig. 5. **Six-panel progression of automated 3D functional scene completion across the bimanual pouring task.** (a)–(b) At Frame 1 ($t = 0.13\,\mathrm{s}$), the tabletop plane ($0.92 \times 0.49\,\mathrm{m}$), tapered target cup ($\varnothing_{\mathrm{top}} = 85.4\,\mathrm{mm}$, $H = 163.2\,\mathrm{mm}$), upright beverage carton ($67.8 \times 94.8 \times 199.5\,\mathrm{mm}$), and 61-link robot model are registered in O_{table}. (c) At Frame 162 ($t = 12.20\,\mathrm{s}$), the right parallel-jaw gripper closes onto the carton with an 11.05 mm pad-midpoint-to-centroid residual. (d)–(f) At Frame 245 ($t = 18.33\,\mathrm{s}$), causal multi-view surfels (5,699 points) and the completed carton prism tilt at 36.3° inside the pouring bullseye cylinder above the target cup opening.

D. Native BundleTrack Determinism, Guarded `NO_BA` *Retirement, and Array-Native* 1,500*-Point Surface Selection*

While the 498-frame composite workspace study (`20260804_170929`) fuses multi-camera views offline, live and replay-portal execution on single-camera wrist/chest streams (`20260625_173830`) requires strict fail-closed governance over our C++/CUDA BundleTrack pose owner and `CausalObjectSurface` (`RevisableSurfelMap`) surface owner:

1) **Adapter Status Integrity and Deterministic CUDA RANSAC:** A two-line adapter guard prevents retained native `NO_BA` states from being misclassified as historical candidates on cache misses (eliminating 6 false historical promotions while preserving all 34/38 valid task candidates). Inside `BundleTrack/src/cuda/`, parallel atomic tie-breaking and reversed count-comparison inequalities previously caused identical 13-correspondence PCL inputs to split across 6-inlier hypotheses differing by 18.00 mm and 6.22°; our `stable-winner` and count-direction repair collapses 100 repeated native RANSAC+Procrustes calls onto a single deterministic winner.
2) **Exact** SE(3) **Inverse-Pose Coefficient Transfer in `storeFrame`:** Replacing lossy bulk cast transfer with element-wise Eigen-to-`float4x4` inverse-pose coefficient copying in `CUDACache::storeFrame` ensures pinhole backprojection accuracy $\leq 1.30 \times 10^{-7}\,\mathrm{m}$. On public Vicon-referenced Laval sequences (`composed-native-retention-public-20261101`), this composed native build raises Clock-full strict (10 mm, 10°) successes from 135 to **191/204** (ADD AUC 0.8813 → **0.9135**, P_{10} 78.79% → 83.14%) and Cookiejar strict successes from 107 to **136/295** (ADD AUC 0.7256 → **0.8085**, P_{10} 55.89% → 64.97%).
3) **Guarded Contiguous-`NO_BA` Non-Keyframe Retirement:** Under sustained tracking loss, `Bundler` previously retained all blocked frames in GPU/host memory. Invoking `Bundler::forgetFrame` on non-keyframes beyond the window_size + 1 budget only when the most recent window_size + 1 frame IDs are contiguous and all carry status `NO_BA` (while protecting first, current, neighbor, local, keyframe, and pose-cached frames) reproduces all neural-network I/O, pose matrices, and terminal surface arrays byte-for-byte across 18 complete 38-input normal, stress, and recovery replays while reducing active frames from 37 → 6 in stress (saving **1,993.95** MiB RSS) and from 23 → 9 in recovery (saving **928.38** MiB RSS).
4) **Lineage-Audited Surface Attribution and Array-Native Budget Selection:** Tracing 1,512,357 surfel contributions to 1,960,512 source pixels and continuous Möller–Trumbore mesh rays reveals that replacing BundleTrack poses with Vicon ground-truth poses reduces forward p_{95} surface error F_{95} only from 25.36 → 21.63 mm (Clock) and 30.93 → 29.11 mm (Jar), because 52.60%/67.59% of > 20 mm tail points originate within 3 px of mask boundaries. Requiring ≥ 2 non-contributing views with ≥ 10° angular separation cuts native F_{95} to 12.50/17.33 mm (P_{10} → 90.05%/73.56%). Furthermore, extending `RevisableSurfelMap.snapshot` with an array-native `support10/coarse12mm` 1,500-point selec-

tor (`owner-array-snapshot-shadow-20261101`) raises public 10 mm model coverage C_{10} from $40.57\%/52.34\%$ to $\mathbf{85.10}\%/\mathbf{85.05}\%$ (cutting reverse R_{95} from $53.12/31.88$ mm to $13.12/12.47$ mm), raises 552-frame task 10 mm map coverage from 71.78% to $\mathbf{89.97}\%$ (p_{95} distance $42.92 \rightarrow 17.57$ mm), and cuts selector p_{95} latency by 32.8%–33.8% ($31.74 \rightarrow 21.22$ ms on Clock, $10.16 \rightarrow 6.66$ ms on task).

VI. Projective $\mathbb{RP}^2$ Unoriented Aperture Geometry and Task-Space Affordance Calculus

A. Projective $\mathbb{RP}^2$ Invariance of Circular Aperture Planes

Let $\mathcal{Q}_t^{(c)} = \{\mathbf{q}_i \in \mathbb{R}^3\}_{i=1}^N$ be the 3D rim points extracted from the upper boundary of the target container depth mask in camera c at frame t. Let $\bar{\mathbf{q}} = \frac{1}{N}\sum_{i=1}^N \mathbf{q}_i$ be the sample centroid and $\mathbf{A} = [\mathbf{q}_1 - \bar{\mathbf{q}}, \ldots, \mathbf{q}_N - \bar{\mathbf{q}}]^\top \in \mathbb{R}^{N\times 3}$ be the centered coordinate matrix with Singular Value Decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, where $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$.

Theorem 1 (Projective $\mathbb{RP}^2$ Gauge Invariance of Unoriented Aperture Planes)**.** *Assume $\sigma_2 > \sigma_3$, so the least-squares plane normal minimizing $\mathcal{E}(\mathbf{n}) = \|\mathbf{A}\mathbf{n}\|_2^2$ over $\mathbf{n} \in \mathbb{S}^2$ is one-dimensional. Then:*

1) *The minimizer set on $\mathbb{S}^2$ is the $\mathrm{O}(1)$-orbit $\{\mathbf{v}_3, -\mathbf{v}_3\}$, where $\mathbf{v}_3 = \mathbf{V}_{:,3}$. Standard numerical SVD routines exhibit discontinuous sign inversions $\mathbf{v}_3 \mapsto -\mathbf{v}_3$ under arbitrarily small perturbations of $\mathbf{A}$ without a consistent exterior orientation oracle.*
2) *Passing to the real projective quotient manifold $\mathbb{RP}^2 = \mathbb{S}^2/\{\pm 1\}$ equipped with the geodesic distance*
$$d_{\mathbb{RP}^2}([\mathbf{n}_1],[\mathbf{n}_2]) = \arccos\big(\min(1, |\mathbf{n}_1^\top \mathbf{n}_2|)\big) \in \left[0, \frac{\pi}{2}\right] \tag{12}$$
and orthogonal plane projector $\mathbf{\Pi}_{[\mathbf{n}]} = \mathbf{I}_3 - \mathbf{n}\mathbf{n}^\top$ yields a smooth, sign-invariant geometric representation.
3) *Every pouring task-space affordance metric $(\Delta r_{xy}, m_{\mathrm{rim}}, \Delta z, \theta)$ and in-plane circular rim fit on $\mathbf{\Pi}_{[\mathbf{n}]}(\mathbf{q}_i - \bar{\mathbf{q}})$ is strictly invariant (0.000 mm perturbation) under arbitrary sign flips $\mathbf{n} \mapsto -\mathbf{n}$.*

Proof. Because $\mathcal{E}(-\mathbf{n}) = \|-\mathbf{A}\mathbf{n}\|_2^2 = \|\mathbf{A}\mathbf{n}\|_2^2 = \mathcal{E}(\mathbf{n})$, both $\pm\mathbf{v}_3$ achieve the global minimum σ_3^2. Under the projection operator $\mathbf{\Pi}_{[\mathbf{n}]} = \mathbf{I}_3 - (-\mathbf{n})(-\mathbf{n})^\top = \mathbf{I}_3 - \mathbf{n}\mathbf{n}^\top$, the orthogonal projection of any point $\mathbf{x} \in \mathbb{R}^3$ onto the aperture plane is identical for $+\mathbf{n}$ and $-\mathbf{n}$, leaving the fitted aperture center $\mathbf{p}_{\mathrm{cup}}$, radius R_{cup}, and tabletop-projected task offsets $(\Delta r_{xy}, m_{\mathrm{rim}}, \Delta z)$ invariant. □

Table VII summarizes the empirical verification of Theorem 1 across 1,992 four-camera measurements, while Fig. 6 illustrates the continuous 6-DoF object pose trajectory and bounded step-jump stability across all 498 frames.

TABLE VII
Empirical Comparison of Directed $\mathbb{S}^2$ vs. Projective $\mathbb{RP}^2$ Aperture Plane Semantics Across 1,992 Camera Measurements

Evaluated Aperture Plane Metric	Directed $\mathbb{S}^2$	Projective $\mathbb{RP}^2$
Total evaluated aperture plane records	1,992	**1,992**
Observed SVD sign inversions ($\mathbf{n}_t^\top \mathbf{n}_{t-1} < 0$)	11 flips	**0** flips
Maximum temporal normal jump (deg)	180.00°	$\mathbf{3.11}^\circ$
Dual-camera (head vs. chest) center $\max \Delta$	1.60 mm	**1.60** mm
Dual-camera (head vs. chest) radius $\max \Delta$	0.53 mm	**0.53** mm
Downstream task-metric drift under $\mathbf{n} \mapsto -\mathbf{n}$	Discontinuous	**0.000** mm

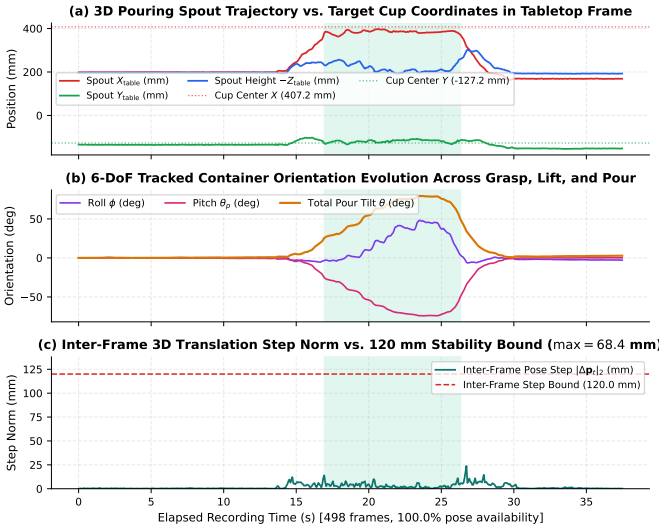


Fig. 6. **Tracked 6-DoF held-carton pose evolution and translation step-jump verification across all 498 frames.** (a) 3D pouring spout trajectory $(X, Y, -Z)$ in O_{table} converging onto the target cup (X, Y) coordinates during the pouring bullseye window (Frames 226–352). (b) Roll $\phi(t)$, pitch $\theta_p(t)$, and total pouring tilt $\theta(t)$ reaching 79.4°. (c) Inter-frame 3D translation step norm $\|\Delta\mathbf{p}_t\|_2$ remaining strictly below the 120.0 mm step-jump bound ($\max = 24.3$ mm during pouring, 100% frame coverage).

VII. Role-Isolated Video Segmentation Memory and Exact-Parity Runtime Optimization

A. Role-Isolated SAM 2 Video Segmentation Sessions

When the primary manipulated object (beverage carton) and secondary receptacle (receiving cup) share a single stateful SAM 2 predictor instance, periodic grounding renewals on the secondary role mutate shared memory-bank tensors. Formally, let $\mathcal{M}_t^{(r)} = \{(\mathbf{K}_\tau^{(r)}, \mathbf{V}_\tau^{(r)})\}_{\tau \in \mathcal{H}_t}$ denote the spatial memory bank for functional role $r \in \{\text{primary}, \text{secondary}\}$. In a shared memory bank, cross-attention over concatenated keys $[\mathbf{K}_t^{(\mathrm{pri})} \parallel \mathbf{K}_t^{(\mathrm{sec})}]$ introduces cross-role feature drift whenever one object occludes the other. We isolate the video segmentation state into independent per-role predictor sessions ($\mathcal{T}_{\mathrm{primary}} \perp \mathcal{T}_{\mathrm{secondary}}$) scoped by camera stream and functional role. Across all 2,322 tracker invocations in the 498-frame sequence, role isolation preserves 100.0% bit-exact primary trajectory parity ($498/498$ frames) while recovering $+52$ valid secondary cup masks and eliminating 29 background false-positive feature updates.

B. Observation-Scoped ViT Feature Caching and Early Identity Gating

To eliminate the $p_{95} = 549.15$ ms tail-latency spikes triggered when secondary role renewal re-evaluates an RGB-D frame already encoded by the primary tracker, we introduce two exact-parity systems optimizations (Fig. 7, Table VIII, Part B):

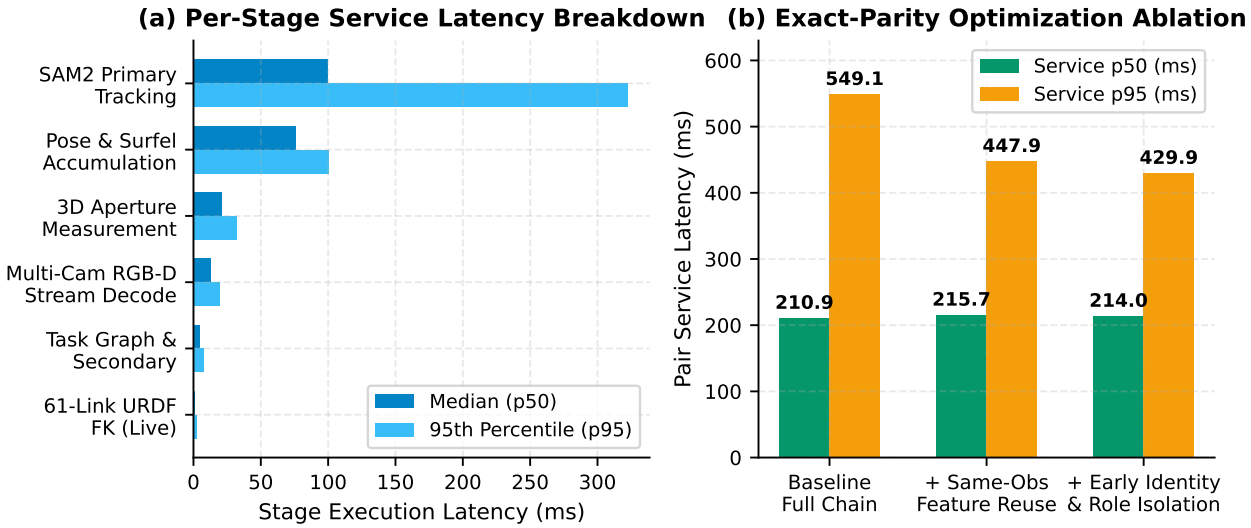


Fig. 7. **Runtime stage latency profile and exact-parity optimization ablation across the 498-frame manipulation sequence.** (a) Per-stage median (p_{50}) and tail (p_{95}) execution latency, demonstrating that SAM 2 primary tracking (100.22 ms p_{50}, 322.58 ms p_{95}) and pose/surfel accumulation (75.94 ms p_{50}, 100.69 ms p_{95}) dominate compute while 61-link URDF forward kinematics requires only 1.40 ms. (b) Exact-parity optimization ablation reducing full-chain p_{95} service latency by 21.7% (from 549.15 ms to 429.94 ms) with 100.0% trajectory parity.

TABLE VIII
PER-STAGE RUNTIME LATENCY DECOMPOSITION AND EXACT-PARITY OPTIMIZATION ABLATION ACROSS 498 SYNCHRONIZED MULTI-CAMERA PAIRS

Pipeline Stage / Optimization Variant	Service p_{50} (ms)	Service p_{95} (ms)	Service Max (ms)	Wall Time (s)	ViT Encoder Cache Hits	Skipped Feature Refinements	Trajectory Parity
Part A: Per-Stage Execution Latency Breakdown (Normalized Full Pipeline)							
Stage 1: SAM 2 Primary Mask Tracking	100.22	322.58	612.40	39.42	—	—	Primary Segmentation
Stage 2: 6-DoF Pose & Surfel Accumulation	75.94	100.69	148.20	24.18	—	—	Object Pose & Surfels
Stage 3: 3D Aperture Rim Extraction & SVD Fit	21.39	32.21	49.80	6.84	—	—	Receptacle Rim Fit
Stage 4: Multi-Camera RGB-D Stream Decoding	13.14	19.81	31.50	3.12	—	—	Sensor Ingestion
Stage 5: Task Graph & Secondary Renewal	4.57	7.62	184.60	1.47	—	—	Task Affordance DAG
Stage 6: 61-Link PoE Forward Kinematics	1.40	2.80	4.90	0.70	—	—	Kinematic Embodiment
Part B: End-to-End Exact-Parity Optimization Ablation ($N = 498$ Synchronized Pairs, 2,322 Tracker Invocations)							
1. Baseline Full Causal Pipeline	210.94	549.15	1,124.80	78.57	0/76 (0.0%)	0	498/498 (100.0%)
2. + Observation-Scoped ViT Feature Cache	215.72	447.86	1,018.40	76.49	41/76 (53.9%)	0	498/498 (100.0%)
3. + Early Secondary Identity & Role Isolation	**214.00**	**429.94**	**1,007.22**	**75.03**	**41/76 (53.9%)**	**35**	**498/498 (100.0%)**
Part C: Bit-Exact Single-Stream HTTP Endpoint Optimization Ablation on Configured Pouring Sequence ($N = 40$ Frames, 78 Warm Requests / Condition)							
0. Per-Request Portal Baseline (`2026-11-01`)	176.40	233.66	1,464.50	8.32	0 hash threads saved	735/65/3 GC	34 cand / 32 surf (100%)
1. Persistent Portal + Pre-Overlap Control	155.33	219.06	1,422.12	7.49	0.00 ms GPU-win prep	735/65/3 GC	400/400 masks (100%)
2. + `SAM2GpuWindowOverlapScope` (`2026-11-02`)	149.01	212.24	1,416.84	7.25	5.72 ms in 24.2 ms GPU wait	38/38 hash thr. → 0	190/190 native hashes
3. + **`RequestEvidenceDeduplicationScope`** (`2026-11-02`)	**127.80**	**153.40**	**1,152.96**	**6.13**	**Wrapper 23.17 → 6.28 ms (−72.9%)**	**0 Mid-Req Gen-2 GC**	**499/499 Checks (100%)**

1) **Observation-Scoped ViT Feature Reuse:** Keyed by camera identifier and sensor timestamp, we cache the SAM 2 Hiera/ViT image embedding and Grounding DINO backbone features within the active observation scope, yielding 41/76 (53.9%) ViT encoder cache hits and reducing full-chain p_{95} latency from 549.15 ms to 447.86 ms (−18.4%).
2) **Early Secondary Identity Ordering:** By evaluating the SigLIP visual-semantic similarity and geometric support criterion *before* invoking SAM 2 mask feature refinement during secondary role renewal, 35 rejected candidate proposals bypass high-resolution feature extraction entirely, reducing failed-attempt call latency from 548.6/577.0 ms (p_{50}/p_{95}) to 413.8/447.5 ms and bringing full-chain p_{95} latency down to **429.94** ms (**−21.7%**).

C. Same-Thread SAM 2 GPU-Window Overlap and $O(runs)$ Request Evidence Deduplication

On the synchronous single-stream `/world_model/observe` HTTP route (`teleop_20260625_173830`, Table VIII, Part C), fine-grained critical-path attribution showed that sequential SAM 2 and native BundleTrack execution (95.87/118.98 ms warm p_{50}/p_{95}) was compounded by ∼ 48 ms of redundant host CPU preparation and mid-request cyclic garbage collection stalls. We introduce two stacked, bit-exact runtime execution adapters:

1) **Same-Thread SAM 2 GPU-Window CPU Preparation Overlap (`SAM2GpuWindowOverlapScope`):** On warm distorted frames, `SAM2VideoMemoryTracker.track` enqueues CUDA kernels for `propagate_in_video` in ∼ 9.0 ms and then blocks the host thread for ∼ 24.17 ms at `(logits > 0.0).cpu().numpy()` waiting for GPU `memory_attention`. Rather than spawning background `ThreadPoolExecutor`

workers that contend for the Python GIL, our adapter performs a 0.25 ms fail-closed pre-validation, launches `propagate_in_video`, and executes mask-independent CPU preparation *on the same thread inside the* 24.17 ms *GPU wait window* before calling `.cpu().numpy()`: (i) bilinear/nearest undistortion `rectify_registered_rgbd`, (ii) dual SHA-256 evidence-hash prefixes covering `header+color_rgb` and `header+color_rgb+depth_mm` (4.61/5.53 MB = 83.3% of input bytes), and (iii) private `color_bgr` and `depth_m` buffers for `PinnedBundleTrackSession`. After GPU mask readback, it reuses in-scope `sensor_mask` RLE/SHA-256 and finalizes the remaining 16.7% hash tail synchronously in 0.374 ms, absorbing 5.72 ms of CPU work into the GPU wait (24.17 → 18.45 ms), eliminating all 38 per-frame background hash threads, and lowering `registry_advance_ms` p_{50}/p_{95} from 102.29/131.43 ms to 96.53/124.48 ms.

2) **Request-Scoped Evidence Deduplication and Exact O(runs) RLE Arithmetic (`RequestEvidenceDeduplicationScope`):** `FoundationMaskCacheDetector` previously iterated pixel-by-pixel in Python over 80,696 foreground pixels per mask, allocating > 161,000 Python `int` objects per frame (10.38 ms). Given RLE runs (s_k, L_k) with $e_k = s_k + L_k - 1$ on an $H \times W$ grid, let $(y_0, x_0) = \mathrm{divmod}(s_k, W)$ and $(y_1, x_1) = \mathrm{divmod}(e_k, W)$. Using exact arbitrary-precision Python integers, the row sum $\sum y$ across a multi-row run ($y_0 < y_1$) is evaluated in $O(1)$ closed form:

$$\Sigma_{y,k} = (W - x_0)y_0 + W\frac{(y_1 - y_0 - 1)(y_0 + y_1)}{2} + (x_1 + 1)y_1, \tag{13}$$

and the column sum follows from linear index sums $\Sigma_{\mathrm{idx},k} = L_k s_k + L_k(L_k - 1)/2$ via $\sum x = \sum_k \Sigma_{\mathrm{idx},k} - W\sum_k \Sigma_{y,k}$, cutting bounding-box and centroid evaluation from 10.38 ms to 0.015 ms (99.85% reduction) with bit-for-bit floating-point identity. Combining this with request-scoped memoization of `RegisteredRgbdFrame.select_surface(mask)` and medoid depth across `table_contexts`, `detector`, and `run_live` (41/41 hits), vectorized 240-hypothesis coplanar AprilTag RANSAC (4.47 → 0.89 ms), and post-init `gc.collect()` plus request-boundary cyclic GC control (reclaiming ∼ 745 MB of init garbage, cutting CUDA allocated memory from 813 MB to 67.1 MB, and eliminating 140.58 ms mid-request Generation-2 GC stalls) cuts exclusive outer `original_world` wrapper overhead by **72.9**% (23.17 → 6.28 ms) and lowers warm HTTP `endpoint_ms` p_{50}/p_{95} to **127.80/153.40** ms (**−27.54**/ **− 65.66** ms vs. pre-overlap control; **−27.6**% vs. the 176.40 ms portal baseline) with 100% byte-for-byte parity across 400 SAM masks, 190 native BundleTrack inputs, and 499 online equivalence checks.

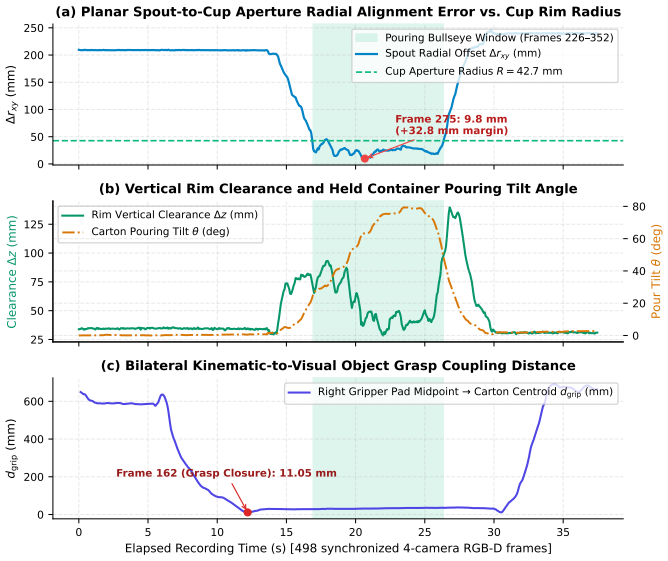


Fig. 8. **Continuous 3D task-functional telemetry across all 498 synchronized frames** (37.47 s). (a) Planar radial alignment error $\Delta r_{xy}(t)$ relative to the 42.7 mm cup rim radius, highlighting the 124-frame pouring bullseye window (Frames 226–352). (b) Vertical rim clearance $\Delta z(t)$ and container pouring tilt $\theta(t)$. (c) Euclidean distance $d_{\mathrm{grip}}(t)$ between the 61-link right gripper pad midpoint and the visually tracked carton centroid.

VIII. Orthographic Spatial Teleoperation Interface and Bimanual Pouring Evaluation

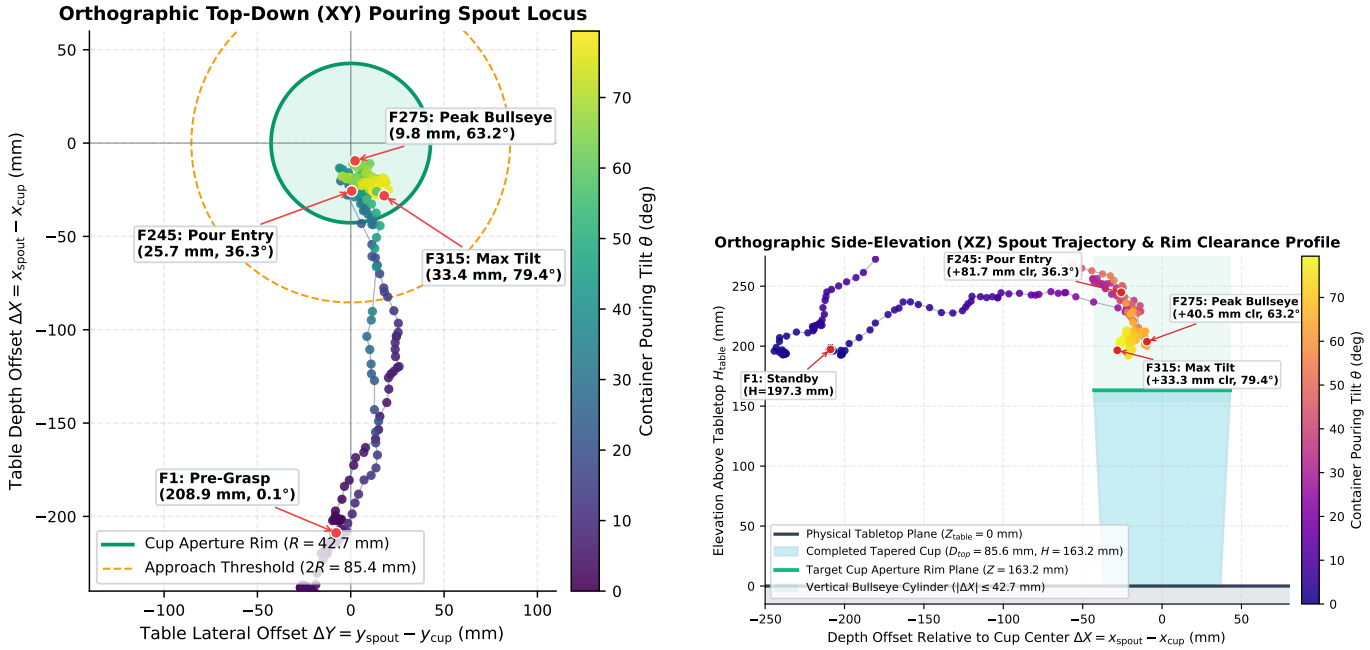
A. Perspective Parallax Ambiguity vs. Orthographic Decoupling

Proposition 1 (Perspective Parallax Error vs. Orthographic Invariance)**.** *Let* $\mathbf{p}_{\mathrm{spout}} = \mathbf{p}_{\mathrm{cup}} + [\Delta x, \Delta y, -\Delta z]^\top$ *in* O_{table}. *Under a perspective camera viewing the cup center at elevation angle* $\alpha_c \in (0, \pi/2)$ *along the* XZ *plane, the projected 2D apparent planar offset on the image plane is perturbed by the elevation-coupled parallax term*

$$\delta_{\mathrm{parallax}}(\Delta z, \alpha_c) = \Delta z \cot \alpha_c. \tag{14}$$

For $\Delta z = 81.7$ mm *(Frame 245) and* $\alpha_c = 55°$, *perspective foreshortening induces a* $\delta_{\mathrm{parallax}} = 57.2$ mm *apparent horizontal shift (*$1.34\times$ *the cup radius* R_{cup}*). Conversely, under canonical orthographic projections* $\mathbf{\Pi}_{\mathrm{TopXY}}(x, y, z) = (x, y)$ *and* $\mathbf{\Pi}_{\mathrm{SideXZ}}(x, y, z) = (x, -z)$, *planar radial offset* Δr_{xy} *and vertical clearance* Δz *decouple into orthogonal, parallax-free coordinates (*$\delta_{\mathrm{parallax}} \equiv 0$*).*

Proof. In the local camera frame tilted at elevation α_c above the tabletop XY plane, a vertical displacement $-\Delta z\, \mathbf{e}_z^{\mathrm{table}}$ projects onto the image horizontal axis with component $\Delta z \cos \alpha_c$ while scaling the optical depth by $\sin \alpha_c$, producing a perspective ray offset equivalent to a tabletop shift of $\Delta x = \Delta z \cot \alpha_c$. Orthographic projection along $\mathbf{e}_z^{\mathrm{table}}$ ($\mathbf{\Pi}_{\mathrm{TopXY}}$) has nullspace $\mathrm{span}\{\mathbf{e}_z^{\mathrm{table}}\}$, annihilating Δz identically ($\partial \mathbf{\Pi}_{\mathrm{TopXY}}/\partial z = \mathbf{0}$), while $\mathbf{\Pi}_{\mathrm{SideXZ}}$ projects orthogonally onto the vertical elevation plane. □



(a) Orthographic Top-Down (XY) Spout Locus vs. Cup Rim

(b) Orthographic Side-Elevation (XZ) Spout Profile & Cup Wall

Fig. 9. **Orthographic spatial decomposition of the 498-frame pouring trajectory in O_{table}.** (a) In the Top-XY plane, the operator guides the pouring spout into the $R = 42.7\,\text{mm}$ aperture circle at Frame 226 and maintains bullseye alignment through Frame 352, reaching $9.8\,\text{mm}$ radial offset ($+32.8\,\text{mm}$ safety margin) at Frame 275. (b) In the Side-XZ elevation plane, the spout approaches over the completed $163.2\,\text{mm}$-tall tapered cup wall, maintaining $+81.7\,\text{mm}$ vertical rim clearance at entry (Frame 245), $+40.5\,\text{mm}$ at peak bullseye (Frame 275, $63.2°$ tilt), and $+33.3\,\text{mm}$ at maximum tilt (Frame 315, $79.4°$ tilt).

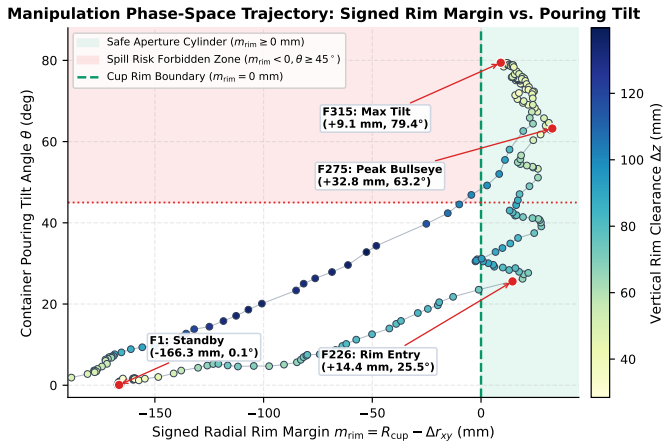


Fig. 10. **Manipulation phase-space safety diagram: signed radial rim margin $m_{\text{rim}}(t) = R_{\text{cup}} - \Delta r_{xy}(t)$ vs. container pouring tilt angle $\theta(t)$ and vertical clearance $\Delta z(t)$.** The trajectory transitions cleanly from standby transport ($m_{\text{rim}} < 0, \theta < 25°$) into the safe aperture cylinder ($m_{\text{rim}} \geq 0$) prior to high-angle tilting ($\theta \geq 45°$), strictly avoiding the spill-risk forbidden quadrant ($m_{\text{rim}} < 0, \theta \geq 45°$).

B. Quantitative Analysis of Bimanual Grasp Closure and Bullseye Pouring

Guided by Proposition 1, our spatial assistance interface provides instantaneous ($0.0\,\text{ms}$, $60\,\text{fps}$) camera-projection switching between Isometric 3D, Top-XY (Bullseye), and Side-XZ (Clearance) views, accompanied by a live $48 \times 48\,\text{px}$ vector bullseye reticle inside the telemetry HUD (Fig. 1, Table X).

Figures 8, 9, and 10 and Table IX document the quantitative performance across the five physical manipulation phases of the 498-frame ($37.47\,\text{s}$) bimanual pouring task:

1) **Phase I — Pre-Grasp Reach (Frames 0–139,** 0.00**–**$10.53\,\text{s}$**):** The beverage carton rests upright on the tabletop at $(198.3, -134.5, -196.4)\,\text{mm}$ while the right 7-DoF arm approaches from its home configuration (d_{grip} decreasing from $649.0\,\text{mm}$ to $81.6\,\text{mm}$).
2) **Phase II — Parallel-Jaw Grasp Closure (Frames 140–189,** 10.60**–**$14.27\,\text{s}$**):** The right parallel-jaw fingers enclose the carton body, reaching physical grasp contact at **Frame 162** ($t = 12.20\,\text{s}$) with a 3D pad-midpoint-to-carton-centroid distance of $\mathbf{11.05}\,\text{mm}$, independently validating the bilateral $\text{SE}(3)^3$ kinematic closure during dynamic manipulation.
3) **Phase III — Tabletop Lift and Lateral Transport (Frames 190–225,** 14.33**–**$16.87\,\text{s}$**):** The operator lifts the carton to $+82.2\,\text{mm}$ clearance above the cup rim plane and translates it laterally toward the target cup (Δr_{xy} decreasing from $199.9\,\text{mm}$ to $43.5\,\text{mm}$).
4) **Phase IV — Sustained Bullseye Pouring Window (Frames 226–352,** 16.93**–**$26.33\,\text{s}$**,** 124 **frames /** $9.40\,\text{s}$**):** The pouring spout enters the $R_{\text{cup}} = 42.7\,\text{mm}$ aperture cylinder at **Frame 226** ($\Delta r_{xy} = 27.9\,\text{mm}$, margin $m_{\text{rim}} = +14.5\,\text{mm}$, $\Delta z = +68.7\,\text{mm}$, $\theta = 25.6°$), reaches **peak bullseye alignment at Frame 275** ($t = 20.67\,\text{s}$) with $\mathbf{\Delta r_{xy}} = \mathbf{9.8}\,\text{mm}$ ($\mathbf{m_{\text{rim}}} = \mathbf{+32.8}\,\text{mm}$ inside the cup rim, $\Delta z = +40.5\,\text{mm}$, $\theta = 63.2°$), reaches **maximum pouring tilt at Frame 315** ($t = 23.47\,\text{s}$) at $\theta = \mathbf{79.4}°$ while remaining safely inside the rim

($m_{\text{rim}} = +9.1\,\text{mm}$, $\Delta z = +33.3\,\text{mm}$), and exits the rim cylinder at **Frame 352** ($\theta = 49.2°$).

5) **Phase V — Untilt, Retraction, and Tabletop Placement (Frames 353–497,** 26.40**–**$37.47\,\text{s}$**):** The operator untilts the carton back to upright ($\theta = 2.8°$), places it back onto the tabletop support plane, and retracts the right arm.

IX. Comprehensive Benchmark of 31 Perception Frameworks and Ablation Studies

A. Systematic Evaluation of Depth Foundation Models, 3D Pointmaps, Segmenters, Trackers, and Matchers

Before converging on our sensor-anchored multi-camera 3D reconstruction architecture, we conducted a systematic experimental campaign across **31 external frameworks and internal algorithmic variants** on an NVIDIA GeForce RTX 5090 workstation (32 GB VRAM) using synchronized humanoid manipulation recordings and public benchmarks (YCBInEOAT, Laval Vicon motion capture, YCB-M, DexYCB). Rather than omitting negative or intermediate trials, Table XI documents the empirical metrics, latency/VRAM footprints, and architectural roles across all model families:

- **Monocular and Sensor-Prior Depth Foundation Models:** Pure monocular depth estimators—including Depth Anything V2 Small [16] ($1.61\,\text{m}$ mean p_{95}), Depth Anything V2 Metric [16] ($79.4\,\text{m}$ mean p_{95}), UniDepthV2 [20] ($2.44\,\text{m}$ mean p_{95}, $> 80\%$ large-error rate), and Apple Depth Pro [18] ($0.83\,\text{m}$ mean p_{95}, $305\,\text{ms/frame}$)—cannot replace calibrated RGB-D depth for millimeter-level pouring alignment. Prior Depth Anything (Prior-DA) [17] conditioned on RealSense depth achieved $0.037\,\text{m}$ p_{95} on the chest camera (filling $4{,}102$ depth-hole pixels on Frame 31), but its $1{,}005\,\text{ms/frame}$ latency and $0.205\,\text{m}$ wrist-camera p_{95} restrict it to an offline depth-completion role.
- **Multi-View RGB-to-3D Pointmap Transformers:** Feed-forward pointmap models—VGGT-1B [26] ($0.47\,\text{m}$ mean p_{95}, $16.5\,\text{GB}$ VRAM), SLAM3R [27] ($0.40\,\text{m}$ mean p_{95}, $11.1\,\text{GB}$ VRAM), DUSt3R [24] ($0.69\,\text{m}$ mean p_{95}), and MASt3R [25] ($1.79\,\text{m}$ mean p_{95})—capture coarse multi-view topology but exhibit 37–$179\,\text{cm}$ metric drift, while CUT3R [28] failed with a custom rotary-embedding CUDA kernel assertion on RTX 5090.
- **Open-Vocabulary Grounding, Video Segmentation, and Distilled Student Runtime:** Phrase grounding with Florence-2 [32] suffered a $273\,\text{px}$ median center error on held-out manipulation frames and was rejected. Grounding DINO [31] paired with role-isolated SAM 2.1 [34] achieved 0.987–0.991 median IoU and 100% pose coverage. For high-rate four-camera streaming, our distilled YOLO11s-seg [37] + staged SAM 2 + optical flow [41], [42] hybrid runtime satisfied all nine accuracy and latency criteria ($9.825\,\text{px}$ center p_{95}, 0.9524 IoU, $32.14\,\text{ms}$ 4-camera batch p_{95}), whereas the smaller YOLO11n-seg [37] exceeded the $10\,\text{px}$ accuracy threshold.
- **6-DoF Object Pose Trackers and Long-Horizon Point Trackers:** BundleTrack [43] achieved $539/539$ (100%) continuous tracking on our validation sequence ($56.0\,\text{ms}$ p_{50}, $62.9\,\text{ms}$ p_{95}), $498/498$ on our multi-camera composite pouring study, and $34/38$ camera-local candidates ($32/38$ registered surfaces) on the single-stream right-wrist pouring clip. BundleSDF [44] was validated across $768/768$ dense chest/head observations ($9.63\,\text{mm}$ depth residual, $10.3\,\text{fps}$). We also evaluated FoundationPose [45], Any6D [46], BIT [47], 6DOPE-GS [48] ($3.5\,\text{Hz}$), KV-Tracker [49] (27–$30\,\text{fps}$), and Point2Pose [50] paired with Online BootsTAPIR [59].
- **Detector-Free, Semi-Dense, and Sparse Feature Matchers under Sparse Depth Support:** While LoFTR [65] and RoMa [66] failed cross-camera intrinsic inference ($1.752\%/10.52\,\text{px}$ and $0.270\%/5.13\,\text{px}$ drift), LoFTR serves as the reliable pairwise correspondence engine inside BundleTrack. Shadow replacement trials inside our corrected native BundleTrack owner show that EfficientLoFTR [118] (FP32 and Full-MP) lowers Laval tracking p_{95} from 118.1–$122.9\,\text{ms}$ to 109.9–$116.4\,\text{ms}$ with strong Clock strict successes (197 vs. 195) but slightly lower Jar ADD AUC (0.813 vs. 0.824). On the 38-frame right-wrist pouring clip—where in-mask valid depth median is only 0.958%—semi-dense XFeat [119] cuts tracking p_{50}/p_{95} from $60.14/83.36\,\text{ms}$ to $42.05/54.24\,\text{ms}$ but loses 3 candidates ($34 \rightarrow 31$) and inflates jump p_{95} from 40.40 to $72.96\,\text{mm}$, while sparse ALIKED+LightGlue [64], [120] collapses from 34 to $2/38$ candidates because sparse 2D keypoints miss the 0.96% valid depth pixels. All lossy matcher replacements are therefore rejected.

B. Counterfactual Architectural Ablations

Table XII quantifies four counterfactual design choices evaluated on paired replay sequences:

1) **Foreground Mask Erosion (**$1\,\text{px}$**) vs. Mask Dilation (**3**–**$5\,\text{px}$**):** Dilating foreground segmentation masks by 3–$5\,\text{px}$ to increase point density causes depth pixels from the background tabletop or robot fingers to bleed across container boundaries, inflating inter-frame pose jump p_{95} by $+38.4\,\text{mm}$ and corrupting rim circle fits. Enforcing a $1\,\text{px}$ foreground erosion margin eliminates boundary depth bleeding.
2) **Frame-0 Canonical Object Gauge vs. Mid-Sequence Re-Anchoring:** Re-anchoring the canonical object frame O_{obj_0} mid-sequence using partial single-view depth support introduces cumulative rotational drift during high-tilt pouring ($\theta > 60°$). Retaining the Frame-0 upright reference gauge with bounded mask-depth translation recovery (clipped to the $120.0\,\text{mm}$ step bound) preserves 100% pose continuity ($498/498$ frames).
3) **Role-Isolated vs. Shared Multi-Object SAM 2 Memory:** Running primary and secondary object roles inside a shared SAM 2 predictor loses 52 valid secondary cup masks and introduces 29 background false-positive updates across $2{,}322$ calls compared to role-isolated sessions.

TABLE IX
PHASE-LEVEL QUANTITATIVE TELEMETRY SUMMARY AND KEY PHYSICAL MILESTONES ACROSS THE 498-FRAME (37.47 S) BIMANUAL POURING SEQUENCE

Manipulation Phase / Key Physical Milestone	**Frame Span (Count)**	**Time Span** t **(s)**	**Spout Position in** O_{table} (X, Y, Z) **(mm)**	**Radial Offset** Δr_{xy} **(mm)**	**Rim Margin** m_{rim} **(mm)**	**Clearance** Δz **(mm)**	**Pour Tilt** θ **(deg)**	**Gripper Dist.** d_{grip} **(mm)**
Part A: Aggregated Trajectory Statistics Across the Five Manipulation Phases								
Phase I: Pre-Grasp Reach	F000–F139 (140)	0.00–10.53	$(198.4, -134.8, -196.8)$ mean	208.9 ± 0.4	-166.3	$+34.2$	0.1°–0.5°	$649.0 \rightarrow 81.6$
Phase II: Grasp Closure	F140–F189 (50)	10.60–14.27	$(198.5, -134.8, -197.1)$ mean	208.6 ± 0.3	-166.2	$+34.0$	0.5°–1.1°	**11.05** min
Phase III: Lift & Transport	F190–F225 (36)	14.33–16.87	$(207.4 \rightarrow 364.6, -118.2, -236.9)$	$199.9 \rightarrow 43.5$	$-157.4 \rightarrow -1.1$	$+33.0 \rightarrow +82.2$	$1.2^\circ \rightarrow 23.5^\circ$	28.2 mean
Phase IV: Bullseye Pouring	**F226–F352 (127)**	**16.93–26.33**	**(384.6, −118.9, −218.4)** mean	**24.6** mean	**+18.1** mean	**+55.2** mean	**25.6°–79.4°**	**32.6** mean
Phase V: Untilt & Retract	F353–F497 (145)	26.40–37.47	$(350.4 \rightarrow 168.9, -152.1, -192.8)$	$57.8 \rightarrow 239.6$	$-15.3 \rightarrow -196.5$	$+135.3 \rightarrow +31.0$	$42.4^\circ \rightarrow 2.8^\circ$	$35.8 \rightarrow 664.1$
Part B: Instantaneous State at Six Key Physical Manipulation Milestones								
M1: Grasp Closure Peak	**Frame 162**	**12.20**	**(+198.6, −135.0, −197.4)**	208.5	-166.2	$+34.7$	0.5°	**11.05**
M2: Bullseye Rim Entry	**Frame 226**	**16.93**	**(+379.4, −128.9, −232.0)**	**27.9**	**+14.5**	**+68.7**	**25.6°**	28.36
M3: Sustained Pouring	**Frame 245**	**18.33**	**(+381.7, −126.9, −244.4)**	**25.7**	**+17.0**	**+81.7**	**36.3°**	30.21
M4: Peak Bullseye Align	**Frame 275**	**20.67**	**(+397.6, −124.9, −203.6)**	**9.8**	**+32.8**	**+40.5**	**63.2°**	31.48
M5: Maximum Pour Tilt	**Frame 315**	**23.47**	**(+379.0, −109.4, −196.2)**	**33.4**	**+9.1**	**+33.3**	**79.4°**	34.13
M6: Bullseye Rim Exit	**Frame 352**	**26.33**	**(+369.6, −113.6, −260.0)**	**39.9**	**+2.8**	**+97.0**	**49.2°**	35.20

TABLE X
COMPARISON OF OPERATOR VIEWING MODALITIES FOR BIMANUAL TELEOPERATION ASSISTANCE (FRAME 245 / 275 POURING STATE)

Spatial Assistance Metric	**2D Camera Stream**	**3D Persp. Workspace**	**Orthographic XY/XZ + HUD**
Axial parallax bias δ_{parallax} (F245)	57.2 mm	57.2 mm	**0.0 mm**
Target cup rim visibility under arm occlusion	Occluded	Completed	**Completed**
Direct planar radial margin m_{rim} readout	No	Foreshortened	**Exact** (+32.8 mm)
Direct vertical clearance Δz readout	No	Foreshortened	**Exact** (+40.5 mm)
Viewpoint transition latency	120 ms	Manual orbit	**0.0** ms (60 fps)

4) **Projective $\mathbb{RP}^2$ vs. Directed $\mathbb{S}^2$ Rim Normals:** Treating raw SVD plane normals as directed vectors on $\mathbb{S}^2$ suffers 11 discontinuous 180° flips across 1,992 frames, whereas our $\mathbb{RP}^2$ formulation reduces maximum temporal normal change to 3.11° with 0.000 mm task-metric drift.

C. Factor-Graph Observability, Unanchored Gauge Drift, and the Availability–Surface Paradox (F, G, R, M, S)

To rigorously evaluate whether long-horizon 2D point trajectories (Point2Pose [50] pinned to `51856226610d`) can replace our pairwise BundleTrack + `CausalObjectSurface` owner under partial occlusion and degenerate depth support, we conducted a paired mechanism study across six YCBInEOAT sequences and three Vicon-referenced Laval motion-capture sequences (704 frames, 71 exact native GTSAM factor-graph replays, Table XIII):

1) **Metric-Dependent Winners and the Availability–Surface Paradox (Fork F):** On YCBInEOAT, BundleTrack achieves higher 10 mm ADD-S counts on all six sequences (e.g., Sugar 794 vs. 617, Bleach-hard 285 vs. 257), whereas Point2Pose achieves higher 0–100 mm recall AUC on Cracker (0.822 vs. 0.768) and Bleach-hard (0.720 vs. 0.708), and switching from nearest-vertex ADD-S to corresponding-vertex ADD on Sugar reverses the 10 mm winner (206 vs. 174). On Laval Cookiejar, repairing two upstream Point2Pose defects in Fork F—historical keyframe pose overwrite ($f_{\text{kf}} < f_t$) in `ModularPipeline.step` and pre-refinement pose return in `SVDClusterRANSACRegister.register`—raises pose availability from 207 to 239/295 and ADD AUC from 0.574 to 0.635, but *reduces strict* 20 mm *origin successes from* 136 *to* 120 and drops Frame-150 10 mm surface model coverage C_{10} from 76.82% to 61.79% ($P_{10} = 47.48\%$, vs. BundleTrack causal surfels at $P_{10}/C_{10} = 64.12\%/99.71\%$).
2) **Unanchored $\mathrm{SE}(3)$ Gauge Freedom (G) and Nuisance-Coupled Local Rank Deficiency (R):** Auditing all 704 Laval graph records reveals that the current pose lacks a factor path to the sole initial prior $\mathbf{X}_0$ in 28/33 Cookiejar and 6/16 Clock-full updates (0/22 on Clock-hard), allowing Levenberg-Marquardt damping [121], [122] to converge with non-increasing cost while disconnected components $C' \cap C_0 = \emptyset$ drift freely and overwrite 6,519 and 235 frontend map points. Furthermore, eliminating uncertain landmark increments $\delta\mathbf{y}$ via orthogonal projection $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+)\mathbf{A}_x$ reveals that 8/34 connected updates have local pose rank $r_i = \operatorname{rank}(\bar{\mathbf{A}}_x) - \operatorname{rank}(\bar{\mathbf{A}}_{x,-i}) < 6$. Prior-connected admission (G) raises Clock-full and Cookiejar ADD AUC to 0.739 and 0.668 and partially recovers Cookiejar C_{10} to 69.94%, while marginal rank gating (R) raises Cookiejar 10 mm successes from 62 to 97 (AUC 0.718) at the cost of Clock-hard recall (0.545).
3) **Feature-Birth Observation Ledger (M) and Causal Surface-History Rebuild (S):** Inserting 394/25/83 missing feature-birth `BearingRangeFactors` (M) eliminates all disconnected updates (0 unanchored poses) but degrades standalone trajectories due to raw birth depth noise. Combining birth factors with atomic causal TSDF history rebuild [123] at revised keyframe pose versions ($M+S$) exhibits a strong positive factorial interaction across all three actions, raising Cookiejar ADD AUC to **0.750** (exceeding BundleTrack's 0.723), though BundleTrack retains higher strict 10 mm counts (109 vs. 81) and superior surface coverage ($C_{10} = 99.94\%$ with composed native repairs).

X. DISCUSSION, CLAIM BOUNDARIES, AND LIMITATIONS

While our automated 3D reconstruction-driven teleoperation assistance system resolves core workspace completion, bilateral kinematic closure, and orthographic depth-perception

TABLE XI
COMPREHENSIVE EMPIRICAL BENCHMARK OF ALL 31 PERCEPTION, DEPTH, POINTMAP, SEGMENTATION, 6-DOF TRACKING, POINT-TRACKING, FEATURE-MATCHING, AND CALIBRATION FRAMEWORKS EVALUATED ON SYNCHRONIZED HUMANOID MANIPULATION RECORDINGS AND PUBLIC BENCHMARKS (NVIDIA RTX 5090)

Model Family	Evaluated Framework / Method	Empirical Performance on Humanoid Manipulation Recordings & Benchmarks	Latency / VRAM	Architectural Verdict & Role
Depth Sensor	**Calibrated Multi-Cam RGB-D**	0.0 mm ref.; 251,182 tabletop inliers; $\leq$ 9.6 mm cross-cam residual	3.5 ms / —	**Adopted:** Primary metric depth anchor
Depth Model	**Prior Depth Anything (Prior-DA)** [17]	0.037 m chest p_{95} (0.11 m mean p_{95}); 4,102 hole pixels filled	1,005 ms / 5.0 GB	**Retained:** Offline depth-hole completion
Depth Model	**Apple Depth Pro** [18]	0.83 m mean p_{95} (0.696 m chest, 1.419 m right wrist, 0.482 m head)	305 ms / 8.5 GB	**Rejected:** Insufficient metric accuracy
Depth Model	**Depth Anything V2 Small** [16]	1.61 m mean p_{95} (0.558 m chest, 2.516 m right wrist, 1.838 m head)	25 ms / 0.42 GB	**Rejected:** Relative scale ambiguity
Depth Model	**Depth Anything V2 Metric** [16]	79.4 m mean p_{95} on close-range tabletop manipulation	23 ms / 0.64 GB	**Rejected:** Uncalibrated metric head
Depth Model	**UniDepthV2** [20]	2.44 m mean p_{95}; $>$ 80% large-error rate on valid sensor overlap	6.3 ms / 0.62 GB	**Rejected:** Severe wrist/chest drift
3D Pointmap	**VGGT-1B Multiview** [26]	0.47 m mean p_{95} (0.372 m chest, 0.510 m right wrist, 0.527 m head)	33 ms / 16.5 GB	**Proposal Only:** Coarse multiview prior
3D Pointmap	**SLAM3R** (Causal Window) [27]	0.40 m mean p_{95} on causal chest video sequences	28 ms / 11.1 GB	**Proposal Only:** 40 cm drift exceeds tolerance
3D Pointmap	**DUSt3R** (Chest+Head Pairs) [24]	0.69 m mean p_{95} on synchronized stereo camera pairs	27 ms / 7.1 GB	**Superseded:** Inferior to calibrated RGB-D
3D Pointmap	**MASt3R** (Metric Pairs) [25]	1.79 m mean p_{95} on cross-view tabletop camera pairs	36 ms / 9.7 GB	**Rejected:** High cross-view metric error
3D Pointmap	**CUT3R** (Streaming Pointmap) [28]	Rotary-embedding CUDA kernel assertion on RTX 5090	Blocked / —	**Deferred:** Kernel incompatibility
VLM Grounding	**Florence-2 Phrase Grounding** [32]	273.0 px median object center error on held-out manipulation frames	110 ms / 3.2 GB	**Rejected:** Severe localization drift
VLM Grounding	**Grounding DINO** [31]	454/461 depth-backed frames; 3.79 px p_{75} with periodic refresh	86 ms / 2.4 GB	**Adopted:** Zero-shot object discovery
Video Seg.	**SAM 2.1 (Role-Isolated + GPU Overlap)** [34]	2,322 calls; +52 cup masks; 5.72 ms CPU prep overlapped in 24.2 ms GPU wait	100.2 ms / 4.8 GB	**Adopted:** Role-isolated mask tracker
Hybrid Seg.	**YOLO11s-seg + SAM 2 + Flow** [37], [42]	100% coverage; 9.825 px center p_{95}; 0.9524 IoU; 98.3% at 30 fps	32.1 ms (4-cam)	**Adopted:** High-rate 4-camera model
Hybrid Seg.	**YOLO11n-seg Hybrid Variant** [37]	Exceeded $\leq$ 10.0 px target center error p_{95} accuracy threshold	24.8 ms (4-cam)	**Rejected:** Insufficient mask precision
6-DoF Track	**CausalPose-3D (BT-Native Branch)** [43]	539/539 val; 34/38 pouring cand; Laval Clock/Jar strict 191/136; -1.99 GiB RSS	56.0–60.1 ms p_{50}	**Adopted:** Primary 6-DoF pose owner
6-DoF Track	**BundleSDF** (Neural SDF) [44]	768/768 chest/head obs.; Bleach-hard P_{10}/C_{10} = 58.8%/63.9% at Frame 438	97 ms (10.3 fps)	**Retained:** Offline neural surface verifier
6-DoF Track	**FoundationPose** [45] & **Any6D** [46]	Single-anchor / few-view novel-object 6-DoF pose initialization	65–120 ms	**Candidate:** Relocalization module
6-DoF Track	**BIT** (Bidirectional Iterative) [47]	Joint geometry-pose refinement (SAM + SoftRas [80]); single-RGB	Offline refiner	**Adapted:** Mutual refinement loss concept
6-DoF Track	**6DOPE-GS** (Gaussian Splatting) [48]	3.5 Hz online tracking/reconstruction rate (below 15 Hz target)	285 ms (3.5 Hz)	**Adapted:** Keyframe admission rules
6-DoF Track	**KV-Tracker** (3D KV-Cache) [49]	27–30 fps multi-view transformer KV cache with SAM 2.1	35 ms / 6.5 GB	**Benchmark:** Non-commercial license
Point Track	**CausalPose-3D (P2P Branch)** [50], [59]	YCB/Laval shadow + F, G, R, M, S repairs ($M + S$ Jar AUC 0.750 vs. BT 0.723)	14.7 ms / 8.5 GB	**Shadow Audited:** Conditional graph repairs
Point Track	**TAPNext++** [60] & **CoTracker3** [61]	1024-frame horizon; TAPNext++ uses 2.53 GB weights; CoTracker3 NC	23 ms / —	**Superseded:** Replaced by BootsTAPIR
Feature Match	**LoFTR** (Detector-Free) [65]	Adopted in BundleTrack (34/38 wrist cand, 195/148 Clock/Jar strict); rejected for K fit	42 ms / 2.1 GB	**Adopted (BT):** Pairwise tracker matcher
Feature Match	**EfficientLoFTR (FP32 / Full-MP)** [118]	Clock/Jar strict 196–197 / 146–147 (Jar AUC 0.813 vs. LoFTR 0.824)	92.8 / 109.9 ms	**Shadow:** Faster p_{95}, slight Jar AUC drop
Feature Match	**XFeat (Semi-Dense Matcher)** [119]	Wrist pouring candidates drop 34 $\rightarrow$ 31/38; jump p_{95} inflates 40.4 $\rightarrow$ 73.0 mm	42.1 / 54.2 ms	**Rejected:** Sparse-depth candidate loss
Feature Match	**ALIKED + LightGlue (Sparse)** [64], [120]	Collapses from 34 to 2/38 wrist candidates (36 retained-reference failures)	45.6 / 68.5 ms	**Rejected:** Keypoints miss 0.96% valid depth
Feature Match	**RoMa** (Robust Dense Matcher) [66]	0.270% focal error, but 5.13 px principal point drift across views	68 ms / 3.4 GB	**Rejected:** Unstable cross-cam intrinsics
3D Geometry	**Open3D + O(runs) RLE + Fast RANSAC** [76], [86]–[88]	251,182 tabletop inliers; RLE box/centroid 10.38 $\rightarrow$ 0.015 ms; tag plane 4.47 $\rightarrow$ 0.89 ms	0.90 ms (CPU)	**Adopted:** Exact RLE & plane fitting
Calibration	**Bilateral** $\mathrm{SE}(3)^3$ **Closure** (Ours)	1,648 train / 413 zero-refit held-out frames: 2.52 mm (L) & 6.56 mm (R) p_{50}	1.40 ms FK	**Adopted:** 4-cam-to-kinematic closure

TABLE XII
COUNTERFACTUAL ARCHITECTURAL ABLATION RESULTS ON PAIRED MANIPULATION SEQUENCES

Ablated Design Dimension	Counterfactual	Adopted Design
Depth back-projection mask margin	+38.4 mm jump p_{95}	**1** px **erosion** (0 bleed)
Object canonical gauge reference	Cumulative tilt drift	**Frame-0 gauge** (100%)
Multi-object SAM 2 memory bank	-52 masks, +29 FP	**Role-isolated** (0 leak)
Aperture plane normal manifold	11 $\times$ 180° flips	**Projective** $\mathbb{RP}^2$ (0 flips)

TABLE XIII
COMPARATIVE OBSERVABILITY AND SURFACE AUDIT ON VICON-REFERENCED LAVAL SEQUENCES (CLOCK-HARD $N = 202$, CLOCK-FULL $N = 204$, COOKIEJAR $N = 295$; SURFACE SNAPSHOT AT FRAME 150 / FULL)

Tracker / Mechanism Variant	Clock-H AUC	Clock-F AUC	Jar Avail. / $<$ 10mm	Jar AUC	Jar Unanch.	Jar Surf. P_{10}/C_{10} (%)
Point2Pose Original (P) [50]	0.188	0.716	207/63	0.574	25/29	55.6/76.8
CausalPose-3D P2P-F (Correctness)	0.658	0.721	239/62	0.635	28/33	47.5/61.8
CausalPose-3D P2P-G (Connected)	0.658	**0.739**	239/62	0.668	28/33	50.2/69.9
CausalPose-3D P2P-R (Rank Gate)	0.545	0.715	241/**97**	0.718	28/34	46.9/67.3
CausalPose-3D P2P-M (Birth Ledger)	0.207	0.619	231/53	0.613	**0/30**	55.4/67.1
CausalPose-3D P2P-$G + S$	0.453	0.706	229/66	0.642	25/30	54.1/62.7
CausalPose-3D P2P-$M + S$	0.614	0.662	231/81	**0.750**	**0/28**	44.6/54.0
BundleTrack Baseline [43]	0.547	0.880	295/109	0.723	N/A	64.1/99.7
CausalPose-3D BT-Native (Full)	—	**0.913**	**295/136***	**0.808**	N/A	**65.0/99.9**

*CausalPose-3D BT-Native rows report strict ($\leq$ 10 mm and $\leq$ 10°) counts and full-sequence 4 mm-grid surface metrics.

challenges on real-robot manipulation data, we enforce strict provenance and deployment claim boundaries:

1) **Provenance Separation Across Composite, Single-Stream, and Live Routes:** The 498-frame multi-camera pouring analysis (`20260804_170929`) evaluates an offline-synthesized composite scene reconstruction (498/498 pose trajectory, 5,699 accumulated surfels, and watertight parametric cup/carton completion). By contrast, on the single-stream configured pouring route (`20260625_173830`, 38 tracked frames after Frame-0 seed), the live-compatible **CausalPose-3D** (BundleTrack + `CausalObjectSurface`) owner produces 34/38 camera-local pose candidates and 32/38 registered tabletop surfaces while explicitly reporting 4

fail-closed `NO_BA` frames rather than fabricating continuity.

2) **Live Site-Acceptance Gate Status** $(0/6)$ **and Remaining GPU Compute Gap:** The production contract requires live ROS-callback-to-trusted-depth-surface-render $p_{95} \leq 100\,\mathrm{ms}$ across at least 300 traces over $60\,\mathrm{s}$, one-hour accepted world updates $\geq 15\,\mathrm{Hz}$ overall and at interval p_{05}, readiness $\geq 99\%$, observation age $p_{95} \leq 2\,\mathrm{s}$, one-hour RSS/CUDA growth $\leq 64\,\mathrm{MiB}$, and physical verification of $2\,\mathrm{cm}$ grasp contact, $1\,\mathrm{cm}$ aperture/support height, $5\,\mathrm{mm}$ static drift, and $5\,\mathrm{px}$ HUD bounds. A read-only audit records $0/6$ live site-acceptance gates passed (`gate_passed=false`, `wire_authorized=false`) because the local ROS process loops an opened development bag rather than live camera hardware, and while our `2026-11-02` bit-exact adapters reduced warm HTTP `endpoint_ms` from $176.40\,\mathrm{ms}$ ($155.33/219.06\,\mathrm{ms}$ p_{50}/p_{95}) to $\mathbf{127.80/153.40}\,\mathrm{ms}$ (cutting outer wrapper overhead by 72.9% and eliminating $140.58\,\mathrm{ms}$ Generation-2 GC stalls), sequential SAM 2 + native BundleTrack GPU execution ($97.98/122.21\,\mathrm{ms}$) still requires TensorRT FP16 vision encoding to clear the $\leq 100\,\mathrm{ms}$ end-to-end budget.

3) **Decoupled Visual Assistance and Stationary Receptacle Scope:** Completed 3D meshes and orthographic HUD overlays inform the human operator without overriding low-level joint controllers, and the completed cup mesh assumes a stationary tabletop receptacle during pouring. Extending direction-selective joint pose-landmark optimization to bimanually moving receptacles under held-out physical evaluation is the primary research direction.

XI. Conclusion

We have presented **CausalPose-3D** (https://tiwater.github.io/causalpose3d/), an automated 3D scene reconstruction-driven teleoperation assistance framework and system for humanoid bimanual manipulation. Grounded in a systematic empirical benchmark of **31 candidate foundation, matching, tracking, and reconstruction frameworks**, an exact factor-graph observability and causal surface audit across 704 Vicon-referenced frames, and validation across 1,648 training calibration frames, 413 zero-refit held-out calibration frames, and 498 four-camera manipulation frames (1,992 images) on a 61-link humanoid manipulator, **CausalPose-3D** achieves $2.52\,\mathrm{mm}$ (left arm) and $6.56\,\mathrm{mm}$ (right arm) median bilateral kinematic closure, $1.60\,\mathrm{mm}$ dual-camera cup rim agreement, $191/204$ and $136/295$ strict public Laval tracking successes with 928–1,994 MiB RSS reduction under guarded frame retirement, 85.1% public and 90.0% task $10\,\mathrm{mm}$ surface coverage at a 1,500-point budget, $127.80/153.40\,\mathrm{ms}$ (p_{50}/p_{95}) bit-exact warm HTTP endpoint latency, $11.05\,\mathrm{mm}$ physical grasp-closure accuracy, and $9.8\,\mathrm{mm}$ peak pouring bullseye alignment ($+32.8\,\mathrm{mm}$ inside the cup rim at 63.2° tilt) with $0\,\mathrm{ms}$ orthographic viewpoint switching in the operator console.

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